Project Portfolio Budgeting as a Markov Decision Process: Environment Design, Baseline Benchmarks, and Optimality Gap Analysis

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Abstract

Allocating a shared capital reserve across a portfolio of simultaneously executing contracted projects is a high-frequency, high-stakes decision for engineering contractors. Classical approaches either assume full foresight or collapse the portfolio to a single project, leaving practitioners with no principled tool for the sequential, multi-project setting under real-world uncertainty. This paper closes that gap by delivering a complete, verified simulation environment that makes the problem tractable for data-driven methods.

We formalise the problem as a Constrained Markov Decision Process in which a portfolio manager allocates budget fractions across N concurrent projects at each discrete period. The state encodes Earned Value Management indicators, S-curve planned progress, milestone payment lookaheads, and a per-project breach-tolerance counter. The transition dynamics reproduce the full FIDIC contract cash-flow cycle: advance payment and recovery, progress-weighted milestone gross payments, retention withholding and release, interest charges on treasury draws, and a two-condition termination gate. The reward signal is the γ-discounted portfolio net cash flow, accumulating to an episode NPV that serves as the primary performance criterion.

The environment is released as a configurable, Gymnasium-compatible package with a SQLite audit ledger that records every allocation, cash-flow event, and breach flag — eliminating the black-box problem inherent to learned policies. To establish a performance baseline and motivate the need for learned agents, we run three deterministic greedy heuristics (Reach-Plan, Catchup-T, and Catchup-Next-T) and a manually computed full-information optimum across four benchmark configurations of increasing portfolio complexity. The results show a systematic and widening gap between the greedy baselines and the optimal solution as portfolio complexity grows, confirming that standard rule-based allocation is insufficient and that a well-defined improvement target exists for reinforcement learning agents. Training such agents and evaluating them against a formal Linear Programming bound are identified as the natural next steps, addressed in subsequent work.

Keywords: project portfolio management; budget allocation under performance uncertainty; constrained Markov decision process; reinforcement learning; FIDIC contracts; earned value management; Gymnasium environment; greedy baseline; optimality gap.

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1 Introduction

1.1 Motivation and Problem Context

Large infrastructure and engineering projects routinely overrun their budgets and miss their schedules (Flyvbjerg et al., 2003; Merrow, 2011). When a portfolio of such projects executes simultaneously under a shared capital commitment, the risks compound: a performance shortfall on one project propagates through the shared budget to all others, and the decision of how much to allocate to each project at each period can determine whether the portfolio succeeds or fails.

Despite this, the formal treatment of portfolio-level budget allocation as a sequential decision problem is largely absent from the project portfolio management literature. Research has concentrated on the *selection* stage — which projects to include — while the *operational governance* stage — how to manage a committed portfolio through to completion — has received comparatively little analytical treatment (Archer and Ghasemzadeh, 1999; Cooper et al., 2001; Turner, 2014). The present work fills this gap by delivering a verified simulation environment that makes the problem tractable for data-driven methods and establishing a performance baseline that quantifies the cost of naive allocation.

We consider a portfolio of projects, each governed by a bilateral contract consistent with FIDIC forms (Fédération Internationale des Ingénieurs-Conseils, 2017), executing concurrently under a shared capital reserve. A portfolio manager exercises one recurring decision at each discrete planning period: how much budget to allocate to each active project. The formal problem is to find the allocation policy that maximises discounted net cash flow across the portfolio horizon while respecting contract mechanics.

1.2 The Project Management Decision Hierarchy

The present contribution operates at the portfolio governance layer of the three-tier project management hierarchy (Project Management Institute, 2017). Portfolio selection — which projects to commit to — is treated as an exogenous input, consistent with the two-stage governance model of Turner (2014). At the governance layer, the portfolio manager does not control project execution directly but allocates budget to each project and observes earned value outcomes. Project termination is not a deliberate agent action but an endogenous environment mechanic triggered by one of two independent conditions: a hard deadline, which fires immediately if the episode period reaches the planned finish plus the contractual finish delay cap; or sustained underperformance, in which a tolerance counter decrements each period the project's actual progress lags behind its planned progress by more than the contractual cap, and termination executes when the counter reaches zero.

Project termination differs from episode termination. A project terminates individually and triggers a settlement cash flow; the episode ends only when every project has either completed or been terminated, when the global horizon is reached, or when the portfolio budget is exhausted with no pending inflows.

1.3 Contract Mechanics and Cash-Flow Structure

The financial interaction between the portfolio manager and each contractor is governed by FIDIC contract forms (Fédération Internationale des Ingénieurs-Conseils, 2017). Three mechanisms determine the cash-flow structure.

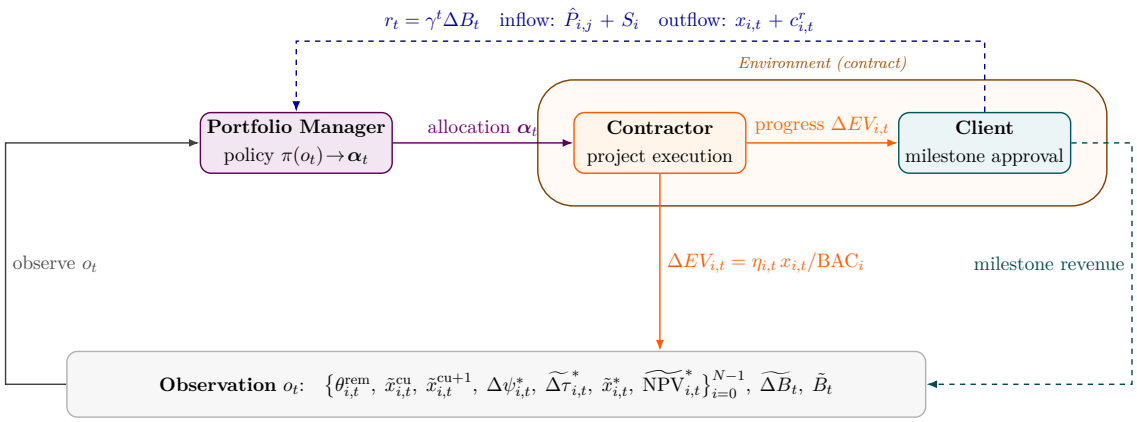


Figure 1: Portfolio governance as a reinforcement learning feedback loop. The portfolio manager commits allocation fractions $\boldsymbol{\alpha}_t$; the contractor converts budget to progress via stochastic efficiency $\eta_{i,t}$; the client certifies milestones and releases net payments $\hat{P}_{i,j}$; the environment returns observation $o_t \in \mathbb{R}^{7N+2}$ and reward r_t. Observation components are defined in full in Table 4.

Milestone-based payment. Payment is contingent on certified progress thresholds. The gross payment at milestone j is a contractually fixed fraction of the contract price. Because payment is contingent on certified progress, the timing of cash inflows depends on the contractor's actual performance and the portfolio manager's allocation decisions. Cash-flow uncertainty in the present model arises from a single source: the stochastic budget-to-progress efficiency $\eta_{i,t}$, which governs how much physical progress a given budget allocation yields in each period. Uncertain productivity induces uncertain milestone timing, which in turn induces uncertain payment arrival — without requiring any separate model of payment delay.

Advance payment and progressive recovery. At project initiation, an advance payment equal to a contractually fixed fraction of the contract price is disbursed to the contractor. It is recovered through percentage deductions from subsequent interim payment certificates, with the commencement threshold and deduction rate specified as project-specific contract parameters (Fédération Internationale des Ingénieurs-Conseils, 2017).

Retention. A fraction of each milestone payment is withheld as a performance bond and released in full upon project completion, creating a structural completion incentive embedded in the payment mechanics.

Earned value management as a performance signal. The EVM framework (Fleming and Koppelman, 2010) provides the diagnostic vocabulary for monitoring contractor performance. The Cost Performance Index and the Earned-Schedule Schedule Performance Index measure cost efficiency and schedule efficiency respectively; the Estimate at Completion uses a composite cost-and-schedule denominator to project total expenditure; and the To-Complete Performance Index measures the efficiency required on remaining work to finish within budget. These four indicators are tracked internally and logged to the audit record each period, but are not directly exposed in the observation vector; they serve as intermediates from which the catchup-allocation and target-milestone

features are derived.

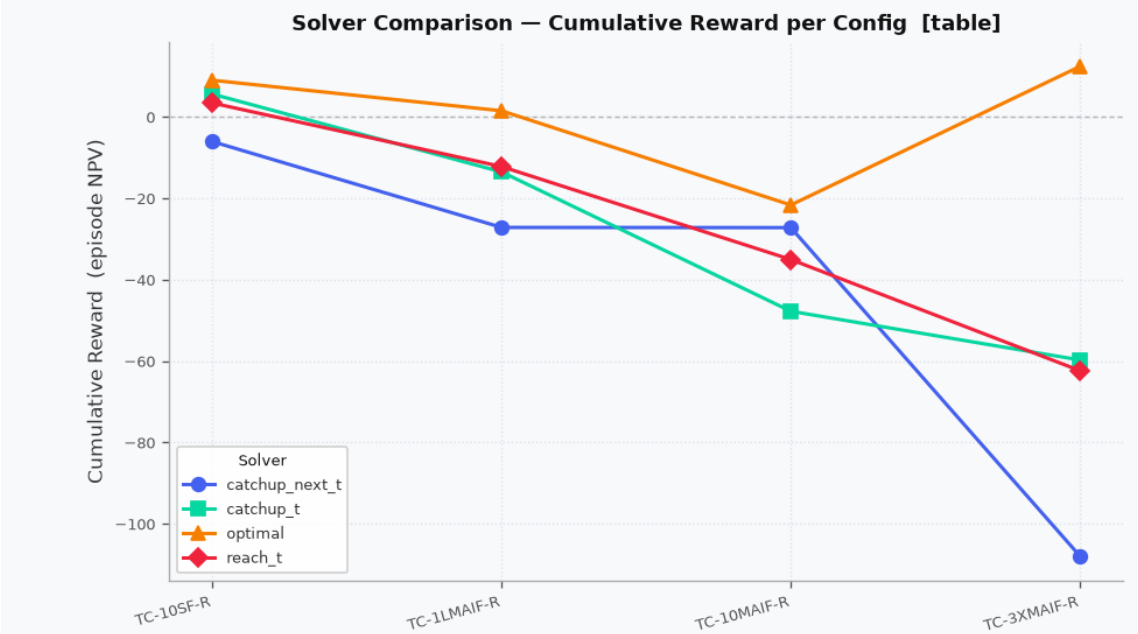
Termination. A distinguishing feature of the proposed environment is that project termination is treated as an endogenous consequence of the manager's own funding decisions, rather than an exogenous shock. In practice, a contractor can lose a project in two qualitatively different ways: gradually, as sustained underfunding allows schedule slippage to persist past a contractually tolerated threshold and through a grace period intended to allow recovery; or abruptly, once the project's finish date overruns its contractually permitted limit regardless of how close it is to catching up. Capturing only one of these failure modes — as prior single-project formulations typically do — understates the manager's true risk exposure. Termination also carries direct financial consequences that must be settled against the portfolio's cash position, since work already paid for and work already completed rarely coincide exactly at the moment a contract is terminated. By embedding both breach conditions, their associated grace mechanism, and the resulting settlement cash flow directly into the transition dynamics, the environment forces any policy to treat the threat of termination as a consequence of its own allocation choices — precisely the incentive structure a real portfolio manager faces.

1.4 The Optimality Gap and the Case for Learned Policies

A central empirical question is whether the problem is hard enough to warrant learned policies at all. To answer this, we run three deterministic greedy heuristics against a manually computed full-information optimum across four benchmark portfolio configurations of increasing complexity.

The **Reach-Plan** heuristic (π^{rp}) allocates exactly enough budget each period to bring every active project to its S-curve planned progress, closing the progress gap entirely. The **Catchup-T** heuristic (π^{cu}) allocates only enough to stay within the contractual progress delay tolerance, minimising expenditure. The **Catchup-Next-T** heuristic ($\pi^{\text{cu}+1}$) is more risk-averse: it pre-emptively allocates enough this period so that the projected delay next period also remains within tolerance. The **full-information optimum** (π^{*}) is computed manually with all stochastic realisations known in advance, serving as an upper-bound reference.

Figure 7 reports cumulative episode NPV for each method across four configurations. On the simplest configuration (single project, fixed efficiency) the greedy baselines come within a small margin of the optimum. As portfolio complexity grows — more projects, staggered starts, interacting cash flows, and milestone sequencing constraints — the gap widens systematically. On the most complex configuration the best greedy baseline trails the optimum by more than 60 NPV units while one baseline degrades to a loss-making outcome. This monotone widening confirms that the allocation problem becomes genuinely difficult at scale and that a well-defined improvement target exists for reinforcement learning agents.



	10SF-R	1LMAIF-R	10MAIF-R	3XMAIF-R
catchup_next_t	-6.07	-27.17	-27.22	-107.96
catchup_t	5.60	-13.53	-47.77	-59.75
optimal	9.00	1.49	-21.66	12.40
reach_t	3.40	-12.21	-35.06	-62.38

Figure 2: Cumulative episode NPV (reward) for three greedy heuristics and the full-information optimum across four benchmark configurations of increasing portfolio complexity. The widening gap as complexity grows motivates the use of learned allocation policies.

1.5 Contributions and Paper Organisation

The contributions of this paper are threefold.

C1. Environment. A configurable, Gymnasium-compatible MDP environment encoding the full FIDIC contract cash-flow cycle: advance payment and progressive recovery, progress-weighted milestone payments, retention withholding and release, treasury-draw interest on negative project cash balances, and a two-condition termination gate with a project-specific cure-period counter. All environment parameters are specified through a configuration dictionary whose entries are probability distributions, allowing practitioners in any project-based sector — construction, IT, R&D, or otherwise — to instantiate domain-specific environments by fitting those distributions to historical company data. Gymnasium API compliance is verified across 51 checks in 10 test sections; the full compatibility certificate is reproduced in Appendix B. The environment ships with a SQLite audit ledger that records every allocation, cash-flow event, and breach flag at each episode step, making policy behaviour fully interpretable.

C2. MDP Formulation. A formal specification of the portfolio budgeting problem as a Markov Decision Process: state space, action space, transition dynamics, reward

signal, and the observation vector presented in closed form, with complete alignment between the mathematical specification and the implemented environment.

C3. Baseline Benchmark and Optimality Gap. Three deterministic greedy heuristics evaluated against a full-information optimum across four benchmark configurations, producing a quantified optimality gap that grows monotonically with portfolio complexity and establishes the performance target for future reinforcement learning agents.

Section 2 reviews the relevant literature across four streams: project management foundations, portfolio management and selection, PPM budgeting and cash-flow modelling, and reinforcement learning for sequential resource allocation. Section 3 develops the MDP formulation formally. Section 5 presents the benchmark evaluation: three deterministic greedy heuristics and a full-information optimum across four portfolio configurations, quantifying the optimality gap that motivates learned agents. Training RL agents on the environment is left as the natural next step, addressed in subsequent work.

2 Literature Review

2.1 Overview and Review Strategy

This section maps the intellectual lineage of the present work across four adjacent knowledge domains: project management (PM) foundations, project portfolio management (PPM) and selection, PPM budgeting and cash-flow modelling, and reinforcement learning (RL) applied to sequential resource-allocation problems. Together these streams define both the theoretical foundation and the gap that motivates the proposed framework. The review follows the PRISMA 2020 reporting guidelines (Page et al., 2021) to ensure transparency and reproducibility in source identification, screening, and inclusion decisions.

2.1.1 Search and Screening Protocol

Literature was retrieved from Google Scholar, Semantic Scholar, and Scopus using a progressively narrowing search strategy. Queries began at the broadest relevant domain (*project management*) and were successively refined through *project portfolio management*, *portfolio budgeting and cash-flow optimisation*, *earned value management under uncertainty*, and *reinforcement learning for project scheduling and resource allocation*. Citation-network traversal — backward and forward snowballing from high-citation anchor papers — was used to recover seminal works not surfaced by keyword search.

Records were screened in three sequential automated stages consistent with the PRISMA 2020 flow diagram (Appendix A). The screening pipeline, together with its archived corpus and output artefacts, is implemented as a reproducible processing chain whose repository location and concept-level mapping are documented in Appendix C.

Stage 1 — Identification and Machine Readability. Candidate PDFs were subjected to automated bibliographic and readability triage. Each document was assessed for text-extraction yield (minimum readable character density across pages), lexical quality (alphabetic-token ratio and mean word length), and sentence-transformer embeddability. Documents failing any criterion were quarantined in a dedicated rejection archive, with a content hash logged to prevent duplicate submissions from re-entering the pipeline.

Stage 2 — Relevance Filtering. Records passing Stage 1 were ranked by a composite relevance score. A semantic similarity component embedded each paper's abstract, introduction, and conclusion using a sentence-transformer model and computed the maximum cosine similarity against twelve thesis-derived query vectors spanning the target knowledge domains. A keyword-scoring component assigned weighted counts of domain-specific terms. The two components were combined with equal weight to yield a combined relevance score in $[0, 1]$. Papers were triaged into *highly relevant* (score ≥ 0.45), *valuable* (score ≥ 0.30), and *irrelevant* (score < 0.30), with the first two cohorts retained for structured extraction.

Stage 3 — Structured Extraction. Each retained paper was subjected to structured knowledge extraction. The abstract, introduction, and conclusion of each PDF were parsed and submitted to a large language model with a fixed schema prompt, producing a per-paper record covering: problem framing, portfolio level, decision type, stochastic cash-flow treatment, contract mechanics, termination-settlement provisions, modelling paradigm, calibration parameters, and thesis-relevance flags. The resulting structured corpus drove an automated synthesis dashboard and table generator, producing the gap-analysis matrix, domain-distribution charts, and the literature map reproduced in Table 1.

The complete pipeline — processing scripts, the structured record corpus, dashboard outputs, and synthesis tables — is archived within the project repository and constitutes a reproducible, machine-auditable literature review corpus; its directory-level organisation is detailed in Appendix C. The PRISMA flow diagram quantifying records at each stage is provided in Appendix A.

2.2 Project Management: Foundational Concepts

The academic study of project management draws on schedule control (CPM/PERT), earned value management (EVM), and risk modelling as its core quantitative pillars (Fleming and Koppelman, 2010; Project Management Institute, 2021). EVM introduced the Cost Performance Index and the Schedule Performance Index as real-time diagnostic signals relating budgeted cost of work performed to actual expenditure and planned schedule — providing the performance vocabulary that the present model inherits directly. The present model adopts the Earned Schedule formulation of the Schedule Performance Index (Lipke, 2003), which continues to degrade past the planned finish date rather than freezing once planned value reaches the budget at completion — a known limitation of the classical ratio in post-planned-finish recovery periods (Lipke, 2003; **?**). The To-Complete Performance Index is included as a forward-looking complement, measuring the cost efficiency required on remaining work to finish within budget. The Estimate at Completion uses a composite cost-and-schedule denominator; Christensen and Heise (1993) established across 155 contracts that this formulation produces the most accurate and stable cost-at-completion forecasts, with reliability increasing beyond the 20% completion mark (Lipke, 2003). Stochastic extensions of EVM (Acebes et al., 2014; Batselier and Vanhoucke, 2017) and Monte Carlo simulation frameworks (Kwak and Ingall, 2007) have modelled performance uncertainty at the project level, establishing that the performance indices follow empirically characterisable distributions over project life and providing the calibration basis for the stochastic efficiency parameter in the present environment.

S-curve analysis, formalised by Kenley and Wilson (2009) for construction cash flows, established that budget consumption follows characteristic sigmoid profiles whose shape parameters are project-type- and contract-form-dependent. The regularised incomplete

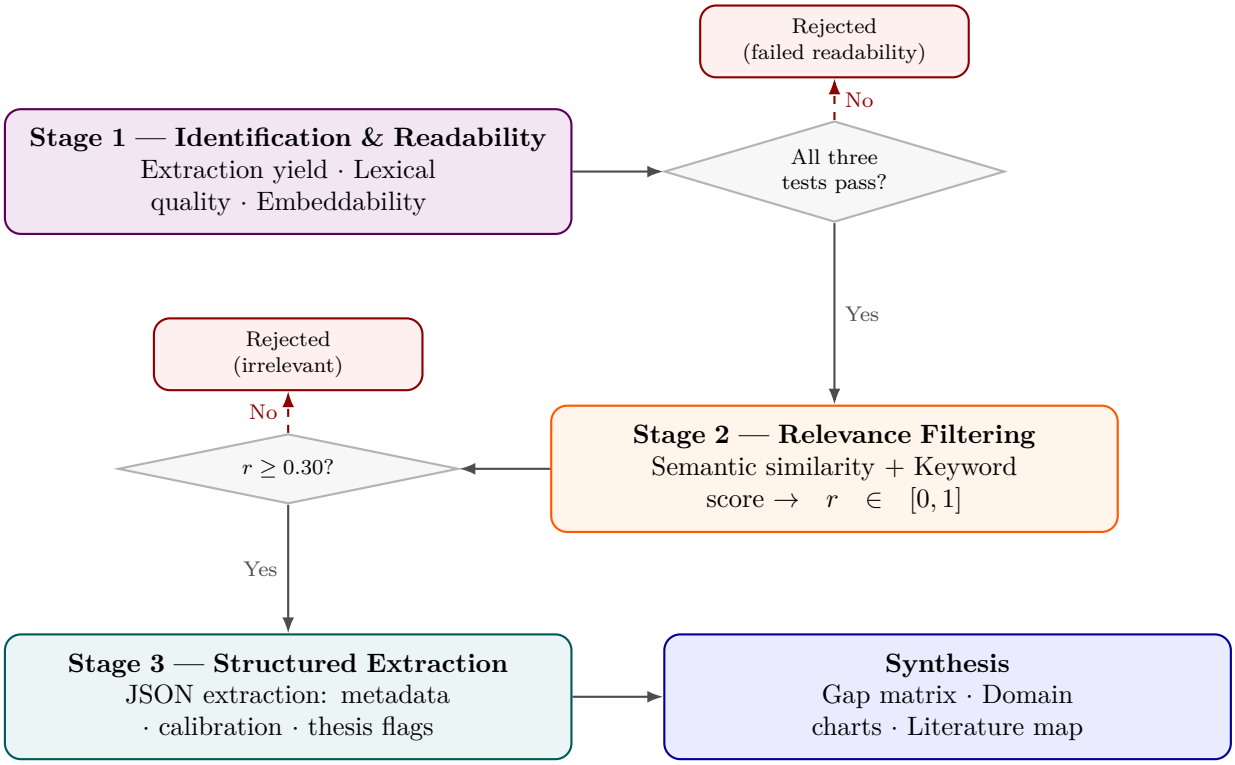


Figure 3: Three-stage PRISMA-aligned screening pipeline, arranged as a serpentine flow. Boxes denote automated processing stages; diamonds denote pass/fail gates; dashed arrows indicate rejection routing. Implementation details and repository locations are provided in Appendix C.

Beta function provides a flexible single-family parameterisation of this profile (Barraza et al., 2004) and is adopted directly in the planned expenditure model of the present formulation.

Large-scale empirical evidence on project performance is provided by Merrow (2011) through the Independent Project Analysis database, and by Flyvbjerg et al. (2003); Flyvbjerg (2017) through infrastructure cost-overrun datasets spanning decades and geographies. These datasets inform both the calibration of uncertainty distributions and the benchmarks against which synthetic environment parameters are validated. Complementary empirical benchmarks are drawn from AACE International and Construction Industry Institute cost-engineering databases (Christensen and Heise, 1993).

Contract governance structures — particularly FIDIC forms (Red, Yellow, and Silver Books) — prescribe the financial interaction between owner and contractor through milestone payment schedules, advance payment and recovery mechanisms, and retention clauses (Fédération Internationale des Ingénieurs-Conseils, 2017). Under FIDIC Sub-Clause 14.2, the advance payment is an interest-free mobilisation loan recovered through percentage deductions from interim payment certificates; the commencement threshold and recovery rate are specified as project-specific particular conditions (Fédération Internationale des Ingénieurs-Conseils, 2017). The present model parameterises this mechanism with project-specific advance payment and recovery rates, accommodating the full range of particular conditions encountered in practice.

2.3 Project Portfolio Management: Selection and Governance

Project portfolio management formalises the problem of allocating an organisation's finite resources across a competing set of candidate projects to maximise strategic value (Archer and Ghasemzadeh, 1999; Cooper et al., 2001). The literature distinguishes two nested decision layers: *portfolio selection* — the binary or mixed-integer optimisation of which projects to include — and *portfolio governance* — the ongoing operational oversight of approved, active projects.

Portfolio selection models have been extensively developed. Markowitz-style mean–variance frameworks adapted from financial portfolio theory (Markowitz, 1952) have been applied to project baskets to trade off expected NPV against variance (Eilat et al., 2006). Multi-criteria decision analysis extensions incorporate strategic alignment, risk ratings, and resource interdependencies (Martinsuo and Lehtonen, 2007; Killen et al., 2008). Metaheuristic approaches address large combinatorial instances (Zhang et al., 2020). Resource-constrained portfolio selection under capital rationing has been studied by Kavadias and Loch (2003), whose no-borrowing constraint is adopted directly in the present formulation.

The present work explicitly adopts the output of portfolio selection as an exogenous input. The selection problem is treated as solved upstream; the focus is entirely on the operational governance layer that receives a committed portfolio and must manage it through execution. This decomposition is consistent with the two-stage governance model of Turner (2014) and the PMI Standard for Portfolio Management (Project Management Institute, 2017), which separates strategic selection from operational execution oversight. In doing so, this work builds directly on the selection literature while advancing the frontier into the governance domain that has received substantially less formal analytical treatment.

2.4 PPM Budgeting, Cash-Flow Modelling, and Uncertainty

Within portfolio governance, budget allocation across active projects over time represents a high-frequency, high-stakes operational decision. Despite its practical centrality, this specific problem has received limited formal treatment relative to portfolio selection. Existing work addresses project-level cash-flow modelling (Kenley and Wilson, 2009; Park and Peña-Mora, 2004), EVM-based forecasting of individual project cost-at-completion (Kim and Ballard, 2002; Acebes et al., 2014), and multi-project resource levelling (Lova et al., 2000; Van Peteghem and Vanhoucke, 2010). These contributions treat budgets as deterministic constraints rather than as dynamic decision variables subject to portfolio-level sequential optimisation.

Cash-flow uncertainty in the present model arises from a single source: the budget-to-progress efficiency, which governs how much physical progress a given budget allocation yields in each period (Acebes et al., 2014; Batselier and Vanhoucke, 2017). Uncertain productivity determines whether and when each milestone threshold is crossed, which in turn determines the timing of the contractually fixed milestone payment. Payment delays between certification and receipt are not modelled as an independent uncertainty source; the performance-driven channel is the structurally primary source of cash-flow timing uncertainty in milestone-payment contracts, and isolating it yields a cleaner identification of the allocation problem without loss of the essential uncertainty framing. Other sources of cash-flow uncertainty — scope changes, external market shocks (Doloi, 2012) — are treated as exogenous.

Stochastic programming (Birge and Louveaux, 2011) provides the canonical OR framework for optimisation under such uncertainty, and its two-stage variant has been applied to project scheduling under resource uncertainty (Lova et al., 2000). However, multi-stage stochastic programs for portfolio-level sequential budget allocation face scenario-tree explosion as the number of projects and periods increases, as formalised in Section 3.4. This intractability motivates the MDP formulation of the present work.

The gap in the literature is precise: no established framework models the portfolio-level budget allocation decision as a sequential stochastic optimisation problem in which a central decision-maker allocates funds across multiple active projects whose cash inflow trajectories are uncertain through stochastic budget-to-progress efficiency, subject to contractual milestone certification windows that bound the speed at which progress converts to cash inflows.

The reward function of the present environment is pure discounted net cash flow; no deviation penalty enters the objective. This is a deliberate design choice: the consequences of sustained deviation — termination and loss of all future milestone payments and retained amounts — are sufficient to make deviation undesirable without a separate penalty term, and including one would introduce an arbitrary scalarisation weight with no natural calibration from contract data (Ng et al., 1999). The discount factor encodes not only time preference but the opportunity cost of tied-up capital in a competitive environment where future projects are foregone while current ones remain incomplete (Damodaran, 2012), giving it a concrete calibration anchor from the firm's observed pipeline redeployment rates or prevailing cost-of-capital benchmarks. A consequence is that a well-performing policy may produce allocation trajectories that concentrate expenditure around milestone certification thresholds, mirroring documented contractor financial strategy in milestone-payment contracts (Kenley and Wilson, 2009; Kaka and Price, 1993; Arditi and Chotibhongs, 2005).

2.5 Contractor Underperformance, Cure Periods, and Termination

When a contractor fails to meet schedule or cost performance thresholds, international EPC contracts prescribe a structured escalation sequence before termination may be exercised. Under FIDIC Sub-Clause 15.1, the client must issue a formal notice of default and allow a cure period — typically 28 days — during which the contractor is obliged to demonstrate measurable recovery before the client may proceed to termination (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008). This cure period is not a discretionary grace; it is a contractual obligation whose omission constitutes wrongful termination. Empirical studies of international EPC disputes confirm that failure to observe the cure period is among the most frequently litigated grounds in contractor termination claims (Ndekugri and Rycroft, 2008; Ennis and Burr, 2021).

The behavioural project management literature has studied the conditions under which organisations fail to terminate underperforming projects at the economically rational point. Escalation of commitment (Staw and Ross, 1987) — the tendency to continue investing in failing projects beyond what expected value justifies — has been documented across construction, IT, and infrastructure programmes. Real options theory (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001) provides the normative counterpart: termination is the exercise of an abandonment option whose value exceeds continuation NPV once performance has deteriorated past a recoverable threshold. Together these streams establish that the decision to persist with or terminate an underperforming project is neither purely contractual nor purely financial — it involves performance signals, contractual obligations, and timing all at once.

The present model integrates all three dimensions into a single environment mechanic governed by two conditions. A progress delay breach fires whenever the lag between planned and actual progress exceeds the contractual tolerance, grounded in the FIDIC engineer's authority to issue a notice when progress falls materially behind programme (Fédération Internationale des Ingénieurs-Conseils, 2017). A hard deadline breach fires unconditionally at the contractually permitted finish overrun and triggers immediate termination regardless of the cure-period counter state. When a progress delay breach is active, the cure-period counter decrements by one; it resets to its initial value only when no breach is present. Termination executes when the counter reaches zero. This mechanic is grounded simultaneously in FIDIC contract law (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008), the real options framing of the abandonment threshold (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001), the EVM literature's validation of performance indices as stable forward signals (Christensen, 1993; Lipke, 2003), and the escalation-of-commitment literature's evidence that observable performance signals must be sustained — not transient — before termination is justified (Staw and Ross, 1987).

The cure-period counter is part of the observable state, giving the policy an anticipatory signal that reflects contractual reality without granting it discretionary control over the termination outcome. From the portfolio manager's perspective, this signal serves a dual function: it flags projects approaching contractual failure while also marking the point at which continued allocation yields diminishing returns relative to the settlement value of work already delivered. A rational manager reading a counter approaching zero in a high-opportunity-cost environment would recognise that the real value of future milestone payments is eroding while the settlement value of work in place is fixed in nominal terms — creating a natural reallocation incentive that the model captures through the interaction of the discount factor and the termination settlement mechanics, without requiring explicit inflation modelling (Dixit and Pindyck, 1994; Fédération Internationale

des Ingénieurs-Conseils, 2017).

2.6 Reinforcement Learning for Sequential Resource Allocation

Reinforcement learning provides a principled framework for sequential decision-making under uncertainty (Sutton and Barto, 2018). The MDP formalisation — state, action, transition, reward — maps naturally onto the portfolio budget allocation problem: the portfolio manager is modelled as a policy that observes the current state, commits an allocation, and receives a discounted net cash-flow reward, with the environment transitioning stochastically according to the realised efficiency draws.

RL has been applied to project scheduling (Peng and Huang, 2021; Liu et al., 2023), resource-constrained project scheduling problems (Tassel et al., 2023), and supply chain inventory management (Giannoccaro and Pontrandolfo, 2002). In each domain, RL demonstrates advantages over classical OR methods when state spaces are large, transition dynamics are stochastic, the problem is non-stationary over the decision horizon, and the horizon length is uncertain at the outset of planning. These are precisely the characteristics of the portfolio budget allocation problem.

The Gymnasium API (Towers et al., 2024) provides the de facto standard interface for RL environment development, enabling direct integration with modern policy optimisation libraries. Domain randomisation (Tobin et al., 2017) — randomising environment parameters across episodes during training — is supported natively through the configuration dictionary, enabling a trained policy to generalise across diverse contract structures and project types without retraining.

Direct application of RL to portfolio-level budget allocation under contractual cash-flow mechanics remains absent from the literature. The present work provides the first MDP formulation of this problem and the first empirical demonstration that deterministic greedy heuristics systematically underperform a full-information optimum as portfolio complexity grows, establishing a quantified improvement target for future learned policies.

2.7 Synthesis: Positioning the Present Contribution

Table 1 maps representative works across the four literature streams against the key modelling dimensions of the present framework, as generated by the automated synthesis pipeline described in Section 2.1.1. Three structural gaps emerge from this mapping.

Gap 1 — Selection versus operational governance. The transition from portfolio *selection* to portfolio *operational governance* has not been modelled as a sequential decision problem. Selection literature assumes away post-selection management; project scheduling literature ignores portfolio interdependencies and the shared capital constraint. No existing work closes both simultaneously.

Gap 2 — Project-level versus portfolio-level uncertainty propagation. Existing stochastic models of project cash-flow uncertainty operate at the individual project level and do not propagate uncertainty upward into a portfolio-level decision architecture governed by a shared budget constraint. The feedback loop between a project's stochastic efficiency trajectory and its portfolio-level funding entitlement remains unmodelled.

Gap 3 — Absence of endogenous termination mechanics. The joint modelling of a progress-delay breach condition tracked against a contractual tolerance, a decrementing cure-period counter governing grace-period exhaustion, an independent hard-deadline breach condition that triggers immediate termination regardless of the cure-period state, and a termination settlement cash flow reconciling earned value against

cumulative payments received, has not been embedded within a portfolio-level sequential decision framework in which budget allocation is the sole control variable.

These gaps are not isolated oversights; they reflect an incomplete progression along the path toward a deployable portfolio-level decision-support system. Figure 4 situates this progression as three sequential steps.

Step 1 — domain grounding — is well established by the literature reviewed above: contract mechanics, EVM performance signals, S-curve expenditure profiles, and portfolio selection models collectively provide the inputs and boundary conditions that any operational governance model requires.

Step 2 — tractable problem formulation — is the gap this paper closes. The contribution is a verified, configurable MDP environment that encodes the full FIDIC cash-flow cycle and a two-condition termination mechanic, released as a Gymnasium-compatible package with a complete SQLite audit ledger. Deterministic greedy heuristics are benchmarked against a full-information optimum, producing a quantified optimality gap that grows monotonically with portfolio complexity and establishes the performance target for future learned agents.

Step 3 — validated deployment, including RL agent training and integration with incumbent project management information systems such as Oracle Primavera P6 — remains future work, contingent on the environment that Step 2 now delivers.

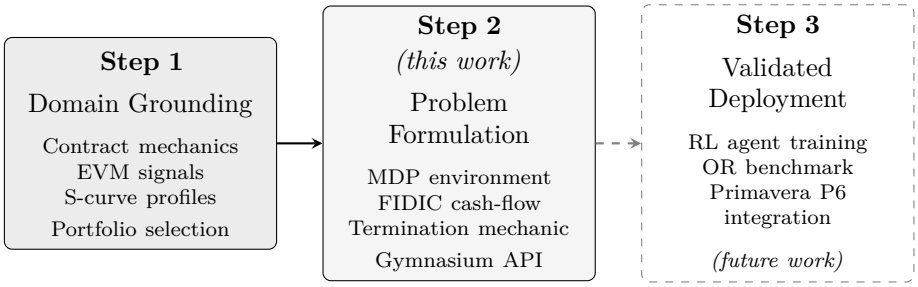


Figure 4: Maturity progression toward a deployable portfolio-level decision-support system. Step 2, completed by the present work, enables Step 3. Dashed border and arrow denote future work.

The present work addresses all three gaps and completes Step 2 of this progression. The Gymnasium-compatible MDP environment — with parameterisable contract mechanics, stochastic budget-to-progress efficiency as the single uncertainty source, and an endogenous two-condition termination mechanic — constitutes the primary methodological contribution. Deterministic greedy heuristics are evaluated against a full-information optimum in Section 5, providing a quantified optimality gap that grows monotonically with portfolio complexity and defines the improvement target for future RL agents.

Table 1: Representative literature mapped against modelling dimensions of the present framework. *Decision type*: S = selection, Sc = scheduling, B = budgeting. ✓ = fully addressed; ○ = partially addressed; – = not addressed.

Work	Portfolio level	Decision type	Sequential decisions	Stochastic cash flows	Contract mechanics	Termination settlement
Yamakawa et al. (2019)	✓	S	–	○	–	–
Kosztyán et al. (2022)	✓	S	–	○	–	–
Touran (2006)	✓	Sc/B	–	○	–	–
Touran (2010)	✓	Sc/B	–	○	–	–
Ghapanchi et al. (2012)	✓	S	–	○	–	–
Pourahmadi et al. (2015)	✓	S	–	○	–	–
El-Sabek and McCabe (2018)	–	B	–	–	✓	–
Beseiso and Kumar (2021)	✓	S	–	○	–	–
Karimi et al. (2021)	✓	S	–	○	–	–
Bahroun et al. (2023)	–	Sc	✓	–	–	–
Ranjbar et al. (2022)	✓	S/Sc	–	–	○	–
Szwarcfiter et al. (2023)	✓	Sc	✓	○	–	–
Asghari et al. (2023)	–	–	✓	○	–	–
Mazhari and Tajik (2024)	✓	S/B	✓	–	–	–
Jiang et al. (2024)	✓	–	✓	○	–	–
Nabot et al. (2025)	✓	Sc	✓	–	✓	✓
Darvish and Sepehri (2025)	✓	B	✓	○	○	✓
Gholami et al. (2025)	✓	Sc	✓	○	–	✓
Kehinde (2025)	✓	B	✓	✓	–	–
Ajibade (2025)	✓	Sc	✓	○	–	–
Huddar and Joshi (2025)	✓	B	✓	○	–	–
Akbari and Shirazi (2026)	✓	B	✓	○	✓	✓
Ebnerasoul et al. (2023)	✓	S	–	○	–	–
Noroozi et al. (2016)	✓	B	–	○	–	–
Jiang et al. (2023)	–	B	✓	–	–	–
Present work	✓	**B**	✓	✓	✓	✓

3 Problem Formulation

This section develops the complete mathematical specification of the portfolio budget allocation problem and its realisation as a Gymnasium-compatible reinforcement learning environment. Section 3.1 states the scope and structural assumptions. Section 3.2 defines the environment primitives: progress dynamics, FIDIC cash-flow mechanics, EVM performance signals, the termination mechanic, and the treasury balance. Section 3.3 consolidates all parameters. Section 3.4 derives the full-foresight upper bound and establishes why the exact stochastic problem is computationally intractable at portfolio scale. Section 3.5 presents the MDP formulation: state space, action space, action projection, transition kernel, reward function, and the observation vector that constitutes the environment's interface to any policy.

3.1 Problem Statement and Scope

The problem is at strategic governance level. A portfolio of N contracted projects is fixed: commitments are signed, start dates are scheduled, and contracts are parameterised consistently with FIDIC Red, Yellow, and Silver Book forms (Fédération Internationale des Ingénieurs-Conseils, 2017). A portfolio manager exercises a single recurring decision at each discrete planning period $t \in \{0, 1, \ldots, T\}$: the monetary budget $x_{i,t} \geq 0$ directed to each active project i. The episode horizon is

$$T = \max_i \left(f_i + \Delta_i^{\max} \right) - 1, \tag{1}$$

where $f_i = s_i + D_i$ is the planned finish and $\Delta_i^{\max}$ is the maximum permitted finish overrun for project i; period $t = 0$ aligns with the earliest planned project start. All outcomes of interest — milestone achievement, cash inflows, schedule adherence, and project termination — flow through the allocation decision. Portfolio selection is treated as an exogenous input (Archer and Ghasemzadeh, 1999).

The formulation rests on four structural assumptions.

(i) *Independent bilateral contracts.* Projects interact only through the shared portfolio cash balance; no resource or workforce sharing occurs, consistent with standard EPC contracting practice.

(ii) *Budget as the sole control.* The contractor's internal resource decisions are absorbed into a stochastic efficiency scalar $\eta_{i,t} \geq 0.01$; the manager observes how much physical progress a budget allocation yields, not how it is achieved. $\eta_{i,t}$ is the sole stochastic primitive and is not observable before $x_{i,t}$ is committed.

(iii) *Discrete monthly periods.* The unit period is one calendar month, reflecting the standard governance cycle. A budget tightness ratio $\kappa = B_0 / \sum_i \text{BAC}_i$ parameterises the liquidity regime.

(iv) *FIDIC contract mechanics.* Advance payment and recovery, milestone revenues, retention withholding and release, and termination settlement are all configurable, accommodating the full range of particular conditions encountered in practice.

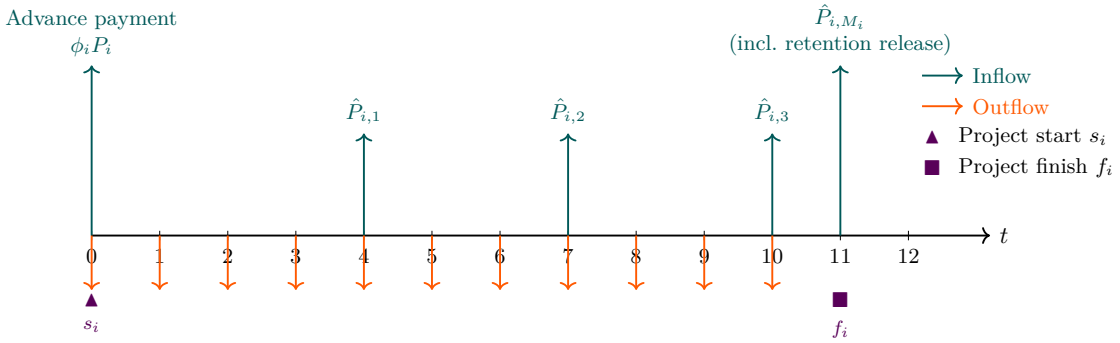


Figure 5: Cash-flow timeline for project i under FIDIC contract mechanics. The budget-to-progress conversion is subject to stochastic efficiency $\eta_{i,t}$, the sole source of uncertainty in the model. Net milestone payments $\hat{P}_{i,j}$ (6) are received at certification; the final payment includes the full retention release.

3.2 Environment Primitives

3.2.1 Progress and Expenditure Dynamics

Let τ_i denote the project-time elapsed for project i since its planned start s_i, incremented by one each period the project is active. Budget $x_{i,t}$ converts to physical progress through the stochastic efficiency scalar $\eta_{i,t}$:

$$EV_{i,t} = \min\Big(1,\ EV_{i,t-1} + \eta_{i,t} \cdot \frac{x_{i,t}}{\mathrm{BAC}_i}\Big), \quad \mathrm{ACWP}_{i,t} = \mathrm{ACWP}_{i,t-1} + x_{i,t}, \tag{2}$$

with $EV_{i,0} = \mathrm{ACWP}_{i,0} = 0$. Here BAC_i is the Budget at Completion and $EV_{i,t} \in [0,1]$ is the fraction of work completed (Budgeted Cost of Work Performed divided by BAC_i). The progress increment $\eta_{i,t}x_{i,t}/\mathrm{BAC}_i$ is realised only after allocation is committed; $\eta_{i,t}$ is drawn from a project-specific distribution whose family and parameters are supplied in the configuration dictionary.

The *planned* cumulative progress follows a normalised S-curve parameterised by Beta shape parameters (a_i, b_i) (Kenley and Wilson, 2009):

$$PV_i(\tau) = \begin{cases} I_{\tau/D_i}(a_i,\ b_i) & \text{if } D_i > 0, \\ 1 & \text{if } D_i = 0, \end{cases} \quad \tau \in [0, D_i], \tag{3}$$

where $I_u(\cdot,\cdot)$ is the regularised incomplete Beta function and D_i is the planned duration. The degenerate case $D_i = 0$ arises when a project's entire planned scope is scheduled within a single period: planned progress is 1 at $\tau = 0$, so the S-curve collapses to a step function and the manager has exactly one period in which to commit the full allocation before the hard deadline fires. $PV_i(\tau_i)$ is the planned progress at the current project-time; $PV_i(\tau_i+1)$ and $PV_i(\tau_i+2)$ serve as one- and two-period lookaheads used in the observation vector and the baseline policies; for $D_i = 0$ both lookaheads return 1.

3.2.2 FIDIC Cash-Flow Mechanics

Contract price and advance payment. The contract price is $P_i = \mathrm{BAC}_i(1 + \pi_i)$, where π_i is the profit margin. If the contract includes an advance payment ($\phi_i > 0$), the portfolio receives at project start ($t = s_i$):

$$\mathrm{Adv}_i = \phi_i \cdot P_i, \tag{4}$$

where ϕ_i is the advance payment rate. When $\phi_i = 0$ no advance is disbursed, the recovery mechanism is inactive, and $A^{\text{rec}}_{i,j} = 0$ for all milestones. The advance is modelled as a synthetic milestone $j = 0$ with progress threshold zero, certified automatically at $t = s_i$, so that $\hat{P}_{i,0} = \text{Adv}_i$ enters V_{s_i} through (7) without special-casing.

Recovery of the advance from subsequent milestone payments follows the FIDIC Sub-Clause 14.2 mechanism (Fédération Internationale des Ingénieurs-Conseils, 2017), governed by two project-specific parameters: a commencement threshold $\psi^{\text{rec}}_i \in [0, 1]$ and a per-payment deduction rate $\delta^{\text{rec}}_i \in [0, 1]$. Let $G^{\text{cum}}_{i,j} = \sum_{k=1}^{j} G_{i,k}$ denote the cumulative gross certified payment through milestone j. The advance recovery deduction at milestone j is:

$$A^{\text{rec}}_{i,j} = \begin{cases} \min\left(\delta^{\text{rec}}_i \cdot G_{i,j},\ \text{Adv}^{\text{rem}}_{i,j}\right) & \text{if } G^{\text{cum}}_{i,j-1} \geq \psi^{\text{rec}}_i \cdot P_i \text{ and } \phi_i > 0, \\ 0 & \text{otherwise,} \end{cases} \tag{5}$$

where $\text{Adv}^{\text{rem}}_{i,j}$ is the advance balance outstanding before milestone j is certified, ensuring the total recovered never exceeds Adv_i. When $\psi^{\text{rec}}_i = 0$, recovery deductions commence from the very first milestone payment.

Milestone certification and net payments. Milestones $j = 1, \ldots, M_i$ are ordered by progress threshold $\psi_{i,1} < \cdots < \psi_{i,M_i} = 1$. Milestone j is certified at the first period t satisfying both $EV_{i,t} \geq \psi_{i,j}$ and $t \geq s_i + \tau^{\text{ms}}_{i,j}$, where $\tau^{\text{ms}}_{i,j}$ is the project-time threshold enforcing the FIDIC Sub-Clause 14.3 payment interval (Fédération Internationale des Ingénieurs-Conseils, 2017). Certification is monotone. The net cash received upon certifying milestone j is:

$$\hat{P}_{i,j} = G_{i,j} - A^{\text{rec}}_{i,j} - R^{\text{held}}_{i,j} + R^{\text{rel}}_{i,j}, \tag{6}$$

where $G_{i,j} = w_{i,j} P_i$ is the gross payment ($w_{i,j}$ is the milestone weight, $\sum_{j=1}^{M_i} w_{i,j} = 1$); $A^{\text{rec}}_{i,j}$ is the advance recovery deduction (5); $R^{\text{held}}_{i,j} = \nu_i G_{i,j}$ is the retention withheld; and $R^{\text{rel}}_{i,j}$ equals the total accumulated retention only at the final milestone, zero otherwise. When $\phi_i > 0$, the full payment identity $\text{Adv}_i + \sum_{j=1}^{M_i} \hat{P}_{i,j} = P_i$ holds exactly; when $\phi_i = 0$ it reduces to $\sum_{j=1}^{M_i} \hat{P}_{i,j} = P_i$. Payments are received in the period of certification.

Period aggregate inflow. The total portfolio inflow at period t is:

$$V_t = \sum_{i=0}^{N-1} \sum_{j=0}^{M_i} \hat{P}_{i,j} \cdot \mathbf{1}[t^{\text{cert}}_{i,j} = t] + \sum_{i \in \mathcal{P}^{\text{term}}_t} S_i, \tag{7}$$

where $t^{\text{cert}}_{i,j}$ is the certification period of milestone j (null if uncertified), $\mathcal{P}^{\text{term}}_t$ collects projects newly terminated at t, and S_i is the termination settlement defined in Section 3.2.4.

3.2.3 EVM Performance Signals

The Earned Value Management framework (Fleming and Koppelman, 2010) provides four diagnostic indices computed each period from observable state. The Earned Schedule $\text{ES}_{i,t}$ is obtained by inverting the planned S-curve at the current actual progress level: it is the project-time $\tau^* \in [0, D_i]$ such that $I_{\tau^*/D_i}(a_i, b_i) = EV_{i,t}$, computed by binary search. The

four indices are:

$$\mathrm{SPI}_{i,t} = \frac{\mathrm{ES}_{i,t}}{\tau_i}, \tag{8}$$

$$\mathrm{CPI}_{i,t} = \frac{EV_{i,t} \cdot \mathrm{BAC}_i}{\mathrm{ACWP}_{i,t}}, \tag{9}$$

$$\mathrm{EAC}_{i,t} = \mathrm{ACWP}_{i,t} + \frac{\mathrm{BAC}_i - EV_{i,t} \cdot \mathrm{BAC}_i}{\mathrm{CPI}_{i,t} \cdot \mathrm{SPI}_{i,t}}, \tag{10}$$

$$\mathrm{TCPI}_{i,t} = \frac{\mathrm{BAC}_i - EV_{i,t} \cdot \mathrm{BAC}_i}{\mathrm{BAC}_i - \mathrm{ACWP}_{i,t}}. \tag{11}$$

The Earned Schedule SPI (8) is used in place of the classical SPI = EV/PV because it continues to degrade after the planned finish date, correctly distinguishing a project one period late from one several periods late (Lipke, 2003). The composite denominator in (10) incorporates both cost and schedule inefficiency into the cost-at-completion forecast (Christensen, 1993); TCPI $>$ 1.10 is widely considered an unrecoverable cost overrun trajectory (Fleming and Koppelman, 2010). All four indices are logged to the audit ledger; they serve as intermediates from which the catchup-allocation and target-milestone observation features are derived, but are not themselves included in the observation vector o_t. Denominators that are effectively zero default to one.

3.2.4 Termination Mechanic

Termination is an endogenous environment mechanic, not an agent action. Two conditions are evaluated independently each period.

Condition 1 Progress delay breach. The signed progress lag at period t is:

$$\delta_{i,t} = PV_i(\tau_i) - EV_{i,t}. \tag{12}$$

A breach flag is raised when the lag exceeds the contractual cap:

$$\mathbb{1}^{\delta}_{i,t} = \mathbf{1}\big[\delta_{i,t} > \delta_i^{\max}\big], \tag{13}$$

where $\delta_i^{\max} \in [0, 1]$ is fixed at commitment, grounded in the FIDIC engineer's authority to issue a notice when progress falls materially behind programme (Fédération Internationale des Ingénieurs-Conseils, 2017).

Condition 2 Hard deadline. A hard deadline fires unconditionally at:

$$\mathbb{1}^{T}_{i,t} = \mathbf{1}\big[t = f_i + \Delta_i^{\max} - 1\big], \tag{14}$$

triggering immediate termination regardless of the breach counter state.

Cure-period counter. Condition 1 activates a cure-period counter modelling the FIDIC Sub-Clause 15.1 recovery interval (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008):

$$\theta^{\mathrm{rem}}_{i,t} = \begin{cases} \max\big(0,\ \theta^{\mathrm{rem}}_{i,t-1} - 1\big) & \text{if } \mathbb{1}^{\delta}_{i,t} = 1, \\ \theta_i & \text{otherwise,} \end{cases} \tag{15}$$

initialised at $\theta^{\text{rem}}_{i,s_i} = \theta_i$. Soft termination executes when $\theta^{\text{rem}}_{i,t} = 0$. Hard termination (14) is evaluated in the early phase (before allocation); Condition 1 and counter updates apply in the late phase (after allocation and progress realisation). The counter $\theta^{\text{rem}}_{i,t}$ is part of the observable state and is directly exposed in the observation vector; the agent influences it indirectly through allocations that drive $EV_{i,t}$ relative to $PV_i(\tau_i)$.

Termination settlement. Upon termination of project i at period t, the net settlement cash flow is:

$$S_i = EV_{i,t} \cdot P_i - \text{CF}^+_i, \tag{16}$$

where CF^+_i is the cumulative inflow received for project i through period t. The first term is the contractor's entitlement for work physically delivered; the second is what has already been collected. $S_i > 0$ when delivered work exceeds prior receipts; $S_i < 0$ imposes a net obligation on the portfolio (advance not yet recovered). In either case S_i is fully determined by contract parameters and the progress level at termination, with no external calibration required.

The state machine governing each project's lifecycle is illustrated in Figure 6.

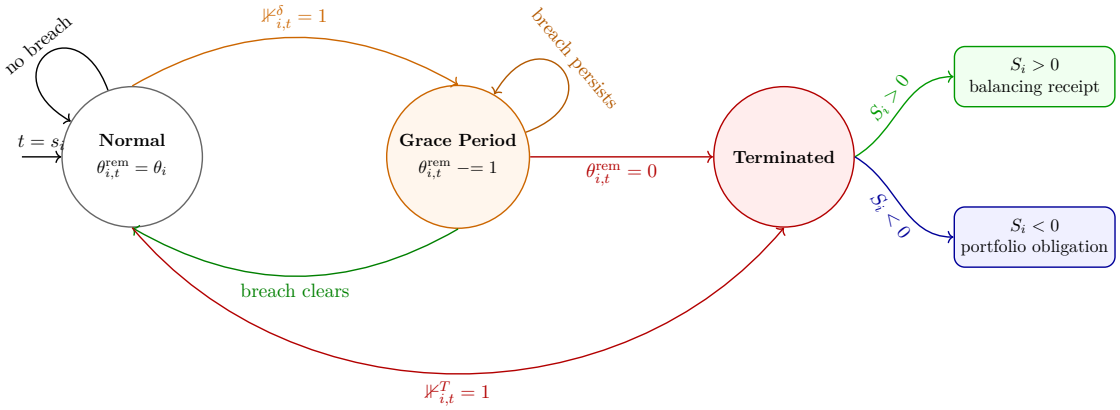


Figure 6: Project lifecycle state machine. Condition 1 (progress delay breach, $\mathbb{1}^{\delta}_{i,t} = 1$) decrements the grace-period counter; Condition 2 (hard deadline, $\mathbb{1}^{T}_{i,t} = 1$) terminates immediately regardless of counter state. Settlement S_i (16) enters portfolio inflows at the termination period.

3.2.5 Treasury Balance and Interest

The net cash flow in period t is $\Delta B_t = V_t - \sum_{i \in \mathcal{P}^{\text{active}}_t} x_{i,t} - \sum_i c^r_{i,t}$, where $c^r_{i,t}$ is the treasury-draw interest charged to project i:

$$c^r_{i,t} = \frac{r}{12} \cdot \max\!\big(0,\ \text{CF}^-_{i,t-1} + x_{i,t} - \text{CF}^+_{i,t-1}\big), \tag{17}$$

where $\text{CF}^-_{i,t}$ and $\text{CF}^+_{i,t}$ are cumulative outflows and inflows for project i through period t, and r is the annual interest rate. Interest accrues when cumulative spending on a project exceeds cumulative receipts — that is, when the project is drawing on treasury reserves rather than being self-funded by milestone inflows. The portfolio balance evolves as:

$$B_{t+1} = B_t + \Delta B_t. \tag{18}$$

When $B_t \leq 0$ and no advance payments are pending, the episode truncates immediately.

3.3 Parameter Reference

Table 3 consolidates all parameters fixed at commitment.

Parameter	Symbol	Domain	Definition
Schedule			
Planned start	s_i	$\mathbb{Z}_{\geq 0}$	Scheduled project start period.
Planned duration	D_i	$\mathbb{Z}_{\geq 0}$	Normalises project-time for S-curve (3); $D_i = 0$ collapses to a step function.
Planned finish	f_i	$\mathbb{Z}_{\geq 0}$	$f_i = s_i + D_i$; defines the hard deadline via (1) and (14).
S-curve shape	$a_i,\ b_i$	$\mathbb{R}^+$	Beta CDF parameters governing $PV_i(\tau)$ (3).
Milestone progress threshold	$\psi_{i,j}$	$(0, 1]$	Minimum $EV_{i,t}$ to certify milestone j.
Milestone timestep threshold	$\tau^{\text{ms}}_{i,j}$	$\mathbb{Z}_{\geq 0}$	Earliest project-time for certification (Fédération Internationale des Ingénieurs-Conseils, 2017).
Progress delay cap	$\delta^{\max}_i$	$[0, 1]$	Maximum lag before breach flag $\not\Vdash^{\delta}_{i,t}$ activates (13).
Finish delay cap	$\Delta^{\max}_i$	$\mathbb{Z}_{>0}$	Extra periods after f_i before hard deadline fires (14).
Termination tolerance	θ_i	$\mathbb{Z}_{>0}$	Grace-period length; models FIDIC Sub-Clause 15.1 (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008).
Contract			
Budget at completion	BAC_i	$\mathbb{R}_{>0}$	Total planned contractor cost.
Profit margin	π_i	$\mathbb{R}_{\geq 0}$	$P_i = \text{BAC}_i(1 + \pi_i)$.
Advance payment rate	ϕ_i	$[0, 0.3]$	$\text{Adv}_i = \phi_i P_i$ at $t = s_i$; $\phi_i = 0$ disables the advance mechanism.
Advance recovery threshold	ψ^{rec}_i	$[0, 1]$	Cumulative certified gross fraction above which recovery deductions commence (5).
Advance recovery rate	δ^{rec}_i	$[0, 1]$	Per-milestone deduction rate to recover Adv_i (5).
Retention rate	ν_i	$[0, 0.1]$	$R^{\text{held}}_{i,j} = \nu_i G_{i,j}$; released in full at $j = M_i$.
Milestone weight	$w_{i,j}$	$[0, 1]$	Fraction of P_i paid at milestone j.
Portfolio			
Initial budget	B_0	$\mathbb{R}_{>0}$	$B_0 = \kappa \sum_i \text{BAC}_i$.
Budget tightness	κ	$\mathbb{R}_{>0}$	Liquidity regime scalar.
Discount factor	γ	$(0, 1]$	Per-period discount (Damodaran, 2012).
Annual interest rate	r	$[0, 1]$	Monthly treasury-draw cost $= r/12$ per (17).
Episode horizon	T	$\mathbb{Z}_{>0}$	Derived from (1).
Stochastic Primitive			
Work efficiency	$\eta_{i,t}$	≥ 0.01	Sole stochastic primitive; drawn per period per project from a configurable distribution; realised after $x_{i,t}$ is committed.

Table 3: Parameters fixed at commitment, grouped by function.

3.4 Full-Foresight Upper Bound and Intractability

3.4.1 Deterministic Oracle Upper Bound

Consider a hypothetical oracle that knows the full efficiency sequence $(\eta_{i,t})_{i,t}$ before making any decision. Under deterministic efficiency, $EV_{i,t}$ is a deterministic function of $\{x_{i,\tau}\}_{\tau \leq t}$, so all milestone certification times and termination periods are computable ex ante. The

oracle maximises:

$$Z^* = \max_{\{x_{i,t} \geq 0\}} \sum_{t=0}^{T} \gamma^t \Delta B_t, \tag{19}$$

subject to $\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t$ for all t, with the balance evolving via (18) and all inflows V_t computed deterministically from the known efficiency trajectory.

Proposition 1 (Full-foresight upper bound)**.** *Let Z^* be the value of (19) on a realised efficiency trajectory. Let Z^π be the discounted cumulative reward achieved by any causal policy π — one that commits $x_{i,t}$ using only information available at time t — on the same trajectory. Then:*

$$\sup_{\pi \in \Pi^{\text{causal}}} \mathbb{E}\big[Z^\pi\big] \leq Z^*. \tag{20}$$

Proof. Fix any realised trajectory. Any allocation sequence produced by a causal policy is also feasible in (19) solved on the same realisations, since feasibility constraints are identical and the oracle optimises over a strictly larger information set. Therefore $Z^\pi \leq Z^*$ pointwise over trajectories, and the inequality is preserved in expectation. □

The gap $Z^* - \mathbb{E}[Z^\pi]$ decomposes into the cost of uncertainty and the quality of the policy. In this work Z^* is computed by hand for each benchmark configuration with full knowledge of the efficiency draws realised during each play session, via the interactive `play.py` interface. These manually obtained solutions constitute the upper-bound reference reported in Section 5.

3.4.2 Intractability of Exact Stochastic Optimisation

The exact stochastic counterpart of (19) requires optimising over policies that map observed history to allocations while respecting the non-anticipativity constraint that $x_{i,t}$ depends only on information available at time t. The canonical solution method is scenario-tree enumeration (Birge and Louveaux, 2011). Discretising $\eta_{i,t}$ to K values per project per period, the tree has

$$|\mathcal{S}| = K^{N \cdot T} \tag{21}$$

leaf nodes. For a moderate portfolio with $N = 3$ projects, $T = 24$ periods, and $K = 5$ discrete scenarios per project per period: $|\mathcal{S}| = 5^{72} \approx 10^{50}$. At $N = 10$ projects the count reaches $5^{240} \approx 10^{167}$, exceeding the number of atoms in the observable universe by more than 80 orders of magnitude. This scaling is intrinsic to the multi-stage stochastic formulation and independent of solver technology. The endogenous active-set dynamics — each termination event structurally changes $\mathcal{P}_t^{\text{active}}$ and bifurcates the scenario tree — introduce a second source of combinatorial growth beyond (21).

This intractability motivates the MDP formulation of Section 3.5: replacing scenario-tree enumeration with sampled trajectory experience, where the computational cost per step is independent of $|\mathcal{S}|$.

3.5 MDP Formulation

3.5.1 Formal MDP Specification

Definition 1 (Portfolio Budget Allocation MDP)**.** The portfolio budget allocation problem is formalised as a finite-horizon Markov Decision Process (Puterman, 1994):

$$\mathcal{M} = \big(\mathcal{S},\ \mathcal{A},\ \mathcal{P},\ R,\ \gamma,\ T\big), \tag{22}$$

with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $\mathcal{P}$, reward function R, discount factor γ, and horizon T.

State space. The state S_t at period t collects, for each project i, the following components:

- actual progress $EV_{i,t} \in [0,1]$,
- cumulative actual cost $\text{ACWP}_{i,t} \in \mathbb{R}_{\geq 0}$,
- EVM indices $\text{SPI}_{i,t}$, $\text{CPI}_{i,t}$, $\text{EAC}_{i,t}$, $\text{TCPI}_{i,t}$,
- project status $\sigma_{i,t} \in \{\texttt{pending}, \texttt{active}, \texttt{completed}, \texttt{terminated}\}$,
- cure-period counter $\theta_{i,t}^{\text{rem}} \in \{0, \ldots, \theta_i\}$,

together with the portfolio cash balance $B_t \in \mathbb{R}_{\geq 0}$. The full state is:

$$S_t \in \mathcal{S} \subseteq [0,1]^N \times \mathbb{R}_{\geq 0}^N \times \mathbb{R}^{4N} \times \Sigma^N \times \mathbb{Z}_{\geq 0}^N \times \mathbb{R}_{\geq 0}, \tag{23}$$

where $\Sigma = \{\texttt{pending}, \texttt{active}, \texttt{completed}, \texttt{terminated}\}$. The state dimension is $9N+1$ per period. All components are observable at decision time. The sole unobservable quantity is $\eta_{i,t}$, realised only after $x_{i,t}$ is committed.

The Markov property holds exactly: S_{t+1} depends on $(S_t, \mathbf{x}_t, \boldsymbol{\eta}_t)$ only, not on the prior history $(S_0, \mathbf{x}_0, \ldots, S_{t-1}, \mathbf{x}_{t-1})$. The full portfolio state S_t is a sufficient statistic for the observable history, making the MDP formulation exact rather than approximate.

Action space. The policy outputs an allocation fraction vector $\boldsymbol{\alpha}_t \in [0,1]^N$ with the interpretation that project i receives fraction $\alpha_{i,t}$ of the available budget. The monetary allocation is $x_{i,t} = \alpha_{i,t} \cdot B_t$.

Action projection. To guarantee feasibility for any raw policy output $\tilde{\boldsymbol{\alpha}}_t \in \mathbb{R}^N$, the environment applies three sequential projection rules before executing the transition:

(i) *Inactive zeroing.* $\tilde{\alpha}_{i,t} \leftarrow 0$ for all i with $\sigma_{i,t} \neq \texttt{active}$.

(ii) *Per-project allocation cap.* The monetary allocation to project i is capped at

$$x_{i,t} \leq \frac{\text{BAC}_i}{\eta_i^{\min}}, \tag{24}$$

where $\eta_i^{\min}$ is the lower bound of project i's efficiency distribution (closed-form for fixed and uniform families; defaulting to 0.01 otherwise). This cap guarantees that a single full allocation always suffices to complete a project regardless of the efficiency draw.

(iii) *Budget projection.* The active sub-vector is projected onto the budget simplex:

$$\boldsymbol{\alpha}_t^{\text{active}} = \frac{\max(\tilde{\boldsymbol{\alpha}}_t^{\text{active}}, \mathbf{0})}{\left\|\max(\tilde{\boldsymbol{\alpha}}_t^{\text{active}}, \mathbf{0})\right\|_1 \vee 1}, \tag{25}$$

where $\vee$ denotes the maximum operator (denominator at least one when all values are non-positive). The resulting monetary allocations $x_{i,t} = \alpha_{i,t} B_t$ satisfy $\sum_i x_{i,t} \leq B_t$ by construction.

This projection is differentiable almost everywhere and preserves gradient flow through policy networks during training (Schulman et al., 2017).

Transition kernel. The stochastic transition $\mathcal{P}(S_{t+1} \mid S_t, \boldsymbol{\alpha}_t)$ is realised by Algorithm 1.

Algorithm 1 Within-period transition kernel at episode step t

Phase 1 Early (pre-allocation)

for each project $i \in \{0, \ldots, N-1\}$ **do**
  if $\sigma_{i,t} = \texttt{active}$ **and** $t \geq f_i + \Delta_i^{\max} - 1$ **then** Termination
    $\sigma_{i,t} \leftarrow \texttt{terminated}$; compute S_i (16); V_t += S_i
  end if
  if $\sigma_{i,t} = \texttt{active}$ **then** Schedule
    Compute $PV_i(\tau_i)$ via (3); update EVM indices (8)–(11) Contract
    for each uncertified milestone j with $EV_{i,t} \geq \psi_{i,j}$ **and** $t \geq s_i + \tau_{i,j}^{\text{ms}}$ **do**
      Certify j; compute $\hat{P}_{i,j}$ (6); V_t += $\hat{P}_{i,j}$
    end for
  end if
end for

Phase 2 Action application

Apply projection (25) to $\tilde{\boldsymbol{\alpha}}_t \Rightarrow \boldsymbol{\alpha}_t$
for each project i with $\sigma_{i,t} = \texttt{active}$ **do**
  $x_{i,t} \leftarrow \alpha_{i,t} \cdot B_t$ Schedule
  Draw $\eta_{i,t}$ from project i's configured distribution
  Update $EV_{i,t}$, $\text{ACWP}_{i,t}$ via (2) Cost
  Compute treasury-draw interest $c_{i,t}^r$ (17)
end for

Phase 3 Late (post-allocation)

for each project i with $\sigma_{i,t} = \texttt{active}$ **do** Schedule
  Recompute EVM indices with post-allocation $EV_{i,t}$ Contract
  for each uncertified milestone j newly satisfying $EV_{i,t} \geq \psi_{i,j}$ **and** $t \geq s_i + \tau_{i,j}^{\text{ms}}$ **do**
    Certify j; compute $\hat{P}_{i,j}$ (6); V_t += $\hat{P}_{i,j}$
  end for Termination
  Evaluate $\not\Vdash_{i,t}^{\delta}$ (13); update $\theta_{i,t}^{\text{rem}}$ via (15)
  if $\theta_{i,t}^{\text{rem}} = 0$ **then**
    $\sigma_{i,t} \leftarrow \texttt{terminated}$; compute S_i (16); V_t += S_i
  else if $EV_{i,t} = 1$ **and** all milestones certified **then**
    $\sigma_{i,t} \leftarrow \texttt{completed}$
  end if
end for

Phase 4 Balance and counter update

Cost

$B_{t+1} \leftarrow B_t + V_t - \sum_i x_{i,t} - \sum_i c_{i,t}^r$ (18) Schedule
$\tau_i \leftarrow \tau_i + 1$ for all i with $\sigma_{i,t} = \texttt{active}$

3.5.2 Reward Function

The per-period reward is the discounted net portfolio cash flow:

$$r_t = \gamma^t \cdot \Delta B_t = \gamma^t \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - \sum_i c^r_{i,t} \right). \tag{26}$$

The policy optimisation objective is:

$$\pi^* = \text{argmax}_\pi \ \mathbb{E}_\pi \left[G = \sum_{t=0}^{T} r_t \right], \tag{27}$$

where the expectation is over the distribution of trajectories induced by π and the efficiency draws $\{\eta_{i,t}\}$.

Structural incentive properties. The reward (26) embeds all performance incentives through the financial structure of the environment, without auxiliary penalty terms:

(i) *Completion incentive.* At project completion, the accumulated retention $\nu_i P_i$ is released together with the final milestone payment, delivering a concentrated inflow that rewards policies which drive projects to full certification.

(ii) *Termination deterrent.* A negative settlement S_i enters V_t as a direct revenue reduction, making termination financially costly without any hand-crafted penalty coefficient. The threat is proportional to the gap between disbursed advances and certified milestone value, incentivising the agent to pace allocations consistent with actual progress.

(iii) *Timing incentive.* Under $\gamma < 1$, certifying milestone j earlier yields $\hat{P}_{i,j}$ at a higher discount weight. A return-maximising policy therefore front-loads allocations to projects where milestone acceleration is feasible (Damodaran, 2012).

(iv) *Treasury-draw deterrent.* Interest cost $c^r_{i,t}$ (17) penalises advance payments that remain unrecovered by certified milestones, discouraging the agent from accepting large advances without managing subsequent progress.

3.5.3 Observation Vector

Rather than exposing the raw state S_t directly, the environment presents a policy-facing observation vector $o_t \in \mathbb{R}^{7N+2}$ constructed from S_t by extracting and normalising the signals most directly relevant to the allocation decision. The raw EVM indices (SPI, CPI, EAC, TCPI) serve as intermediates and are not included in o_t; instead, they are aggregated into two groups of actionable per-project features.

Catchup signals. These encode the minimum spend to avoid a contractual breach, computed from the S-curve lookahead. At project-time τ_i, the minimum cumulative progress required by end of the current period to stay within the delay cap is:

$$\underline{EV}_{i,t} = \max\big(0,\ PV_i(\tau_i + 1) - \delta_i^{\max}\big). \tag{28}$$

The catchup allocation (current period) is:

$$x^{\text{cu}}_{i,t} = \max\big(0,\ \underline{EV}_{i,t} - EV_{i,t}\big) \cdot \text{BAC}_i. \tag{29}$$

The next-period lookahead extends one step further using $PV_i(\tau_i{+}2)$ in place of $PV_i(\tau_i{+}1)$:

$$x^{\text{cu}+1}_{i,t} = \max\big(0,\ PV_i(\tau_i + 2) - \delta_i^{\max} - EV_{i,t}\big) \cdot \text{BAC}_i. \tag{30}$$

Target milestone signals. The target milestone $j^*_{i,t}$ is the next uncertified milestone for project i at period t. Four signals are derived from it:

$$\Delta\psi^*_{i,t} = \max\big(0,\ \psi_{i,j^*} - EV_{i,t}\big), \tag{31}$$

$$\Delta\tau^*_{i,t} = \big(s_i + \tau^{\text{ms}}_{i,j^*}\big) - t, \tag{32}$$

$$x^*_{i,t} = \Delta\psi^*_{i,t} \cdot \text{BAC}_i, \tag{33}$$

$$\text{NPV}^*_{i,t} = \frac{\hat{P}^*_{i,t}}{\gamma^{\max(0,\,\Delta\tau^*_{i,t})}} - x^*_{i,t}, \tag{34}$$

where $\hat{P}^*_{i,t} = \hat{P}_{i,j^*}$ is the net payment of the target milestone. Equation (34) discounts the expected payment back to the current period and subtracts the investment required, giving the net present value of unlocking the next milestone under the assumption that the payment arrives at the earliest eligible period.

Observation construction. The observation vector is the flattened, clipped concatenation:

$$o_t = \Big(\big\{\theta^{\text{rem}}_{i,t},\ \tilde{x}^{\text{cu}}_{i,t},\ \tilde{x}^{\text{cu}+1}_{i,t},\ \Delta\psi^*_{i,t},\ \widetilde{\Delta\tau}^{\,*}_{i,t},\ \tilde{x}^*_{i,t},\ \widetilde{\text{NPV}}^{\,*}_{i,t}\big\}_{i=0}^{N-1},\ \widetilde{\Delta B}_t,\ \tilde{B}_t\Big), \tag{35}$$

where tilde notation denotes normalisation by B_0 (for monetary quantities), by T (for timestep gaps), or by $\max_i |\text{NPV}^*_{i,t}|$ (for the NPV component), followed by clipping to the ranges listed in Table 4. For pending or completed projects, all per-project fields are set to zero. The portfolio scalars $\widetilde{\Delta B}_t$ and $\tilde{B}_t$ are the last-period net cash flow and the current available budget, both normalised by B_0.

Index	Symbol	Clip	Description
$7i$	$\theta^{\text{rem}}_{i,t}$	$[0, \theta_i]$	Cure-period tolerance remaining (15)
$7i+1$	$\tilde{x}^{\text{cu}}_{i,t}$	$[0, 1]$	Current-period catchup allocation $x^{\text{cu}}_{i,t}$ / B_0 (29)
$7i+2$	$\tilde{x}^{\text{cu}+1}_{i,t}$	$[0, 1]$	Next-period catchup allocation $x^{\text{cu}+1}_{i,t}$ / B_0 (30)
$7i+3$	$\Delta\psi^*_{i,t}$	$[0, 1]$	Target milestone progress gap (31)
$7i+4$	$\widetilde{\Delta\tau}^{\,*}_{i,t}$	$[-1, 1]$	Target milestone timestep gap $\Delta\tau^*_{i,t}$ / T (32)
$7i+5$	$\tilde{x}^*_{i,t}$	$[0, 1]$	Target required allocation $x^*_{i,t}$ / B_0 (33)
$7i+6$	$\widetilde{\text{NPV}}^{\,*}_{i,t}$	$[-1, 1]$	Target milestone NPV, normalised by $\max_i \lvert\text{NPV}^*_{i,t}\rvert$ (34)
$7N$	$\widetilde{\Delta B}_t$	$[-2, 2]$	Last-period net cash flow ΔB_{t-1} / B_0
$7N+1$	$\tilde{B}_t$	$[0, 2]$	Available budget B_t / B_0

Table 4: Observation vector o_t (35): components, normalisation, and clipping bounds. Per-project block i occupies indices $7i$ through $7i+6$.

3.5.4 Gymnasium-Compatible Implementation

The MDP $\mathcal{M}$ is implemented as `PortfolioBudgetingEnv`, a configurable Gymnasium-compatible environment (Towers et al., 2024). The environment exposes the standard `reset()` / `step()` interface:

- `observation_space`: Box($7N + 2$) with per-component bounds as in Table 4;
- `action_space`: Box($[0, 1]^N$), raw allocation fractions of B_t.

All parameters (Table 3) are passed as a configuration dictionary at initialisation; every entry is a probability distribution specified by a family name and up to four parameters (fixed, uniform, Beta, triangular, lognormal, truncated normal, or per-project custom ordinal sequence). This design allows any parameter to be held constant for controlled experiments or drawn fresh each episode for domain randomisation (Tobin et al., 2017). The horizon T is derived automatically from (1); N is sampled at each `reset()` if the portfolio size is configured as a distribution. Episodes terminate when all projects reach `completed` or `terminated` status, or when $t \geq T$. A SQLite audit ledger records every allocation, cash-flow event, and breach flag at each step, enabling full interpretability of any policy's behaviour.

3.5.5 Deterministic Benchmark Policies

Three deterministic benchmark policies are defined over o_t to provide performance baselines and quantify the gap to the full-foresight upper bound. Each reads directly from the catchup signals in o_t and passes through the same three-step projection (25) as any other policy.

Reach-Plan (π^{rp}). Allocate exactly enough to bring each active project to its on-plan progress $PV_i(\tau_i + 1)$:

$$x_{i,t}^{\text{rp}} = \max\big(0,\ PV_i(\tau_i + 1) - EV_{i,t}\big) \cdot \text{BAC}_i. \tag{36}$$

This closes the progress gap fully each period but ignores milestone timing, contract profitability, and budget scarcity.

Catchup-T (π^{cu}). Allocate the minimum required to avoid a breach this period: $x_{i,t}^{\text{cu}}$ from (29). This is the least-cost strategy that avoids activating the breach counter each period.

Catchup-Next-T ($\pi^{\text{cu+1}}$). One-period lookahead: allocate $x_{i,t}^{\text{cu+1}}$ from (30), preventing both current- and next-period breaches at the cost of higher early expenditure and lower discounted returns.

The three policies span a range from aggressive (Reach-Plan, highest spend) to conservative (Catchup-T, minimum spend), providing a meaningful baseline interval. Their performance across four benchmark configurations is analysed in Section 5.

4 Evaluation

5 Computational Experiments and Environment Validation

This section validates the environment mechanics and quantifies the optimality gap available to future learned policies. Three deterministic greedy heuristics and a manually computed full-information optimum are evaluated across four benchmark portfolio configurations of increasing structural complexity. The configurations are designed so that each one isolates a specific mechanic of the environment — progress dynamics, advance payment timing, termination risk, multi-project coordination — enabling precise attribution of each heuristic's failure mode. All episodes are executed through the Gymnasium environment described in Section 3, with the SQLite audit ledger providing a complete per-period record of allocations, cash-flow events, and breach flags.

5.1 Shared Parameters

All four configurations share the contract and environment parameters in Table 5, fixed across every episode and every solver so that observed performance differences are attributable solely to configuration structure and solver policy.

Parameter	Symbol	Value
Retention rate	ν_i	0.05
Progress delay cap	$\delta_i^{\max}$	0.15
Finish delay cap	$\Delta_i^{\max}$	2 periods
Termination tolerance	θ_i	2 breach-periods
Annual interest rate	r	0.12
Per-period deficit rate	$r/12$	0.01
Budget tightness	κ	3.0
Profit margin	π_i	0.20
Discount factor	γ	0.95
S-curve shape	$(a_i,\, b_i)$	$(1.5,\ 3.0)$

Table 5: Parameters shared across all benchmark configurations. Within each milestone payment, advance recovery is applied first to the gross payment; retention is deducted from the same gross payment independently. Accumulated retention is released in full at the final milestone.

Duration convention. A project with planned duration D_i executes over project-timesteps $\tau_i \in \{0, 1, \ldots, D_i\}$, giving $D_i + 1$ active periods. With finish delay cap $\Delta_i^{\max} = 2$, the hard deadline falls at episode period $f_i + 2$, and each efficiency sequence has length $D_i + 3$ entries covering all possible active periods.

Deduction order. Advance recovery is applied first to the gross milestone payment; retention is deducted from the same gross payment independently and in parallel. At the final milestone, any advance not yet recovered is cleared in full and all accumulated retention is released.

5.2 Benchmark Configurations

Table 6 summarises the four configurations. Each uses a fixed custom-ordinal efficiency sequence ($\eta_{i,t}$ set to a deterministic per-project list) under seed = 0, ensuring exact reproducibility.

TC-10SF-R — Isolated atom. One project, one non-advance milestone, no advance payment, planned duration $D_i = 0$. The project has a single planned action at $\tau_i = 0$ and the hard deadline falls at episode period 2. Efficiency sequence: [0.91, 1.17, 0.83]. Contract price $P_i = 120$; the single milestone carries weight $w = 1.0$, gross payment = 120; retention of $\nu_i \times 120 = 6$ is withheld and immediately released at the same (final) milestone, giving net payment = 120 with no advance deductions. This configuration provides the cleanest possible baseline, testing the fundamental progress-to-cash-flow loop with no confounding payment mechanics.

Config	N	BAC_i	D_i	s_i	M_i	ϕ_i	ψ_i^{trig}	**Primary mechanic**
TC-10SF-R	1	100	0	0	1	0.00	—	Isolated atom: single action, single milestone
TC-1LMAIF-R	1	100	3	2	3	0.20	0.80	Delayed start, progress-triggered advance, high reco
TC-10MAIF-R	1	100	3	0	3	0.20	0.00	Diminishing efficiency, termination risk
TC-3XMAIF-R	3	100/200/300	3	0/6/7	3	0.20	0.00	Multi-project, gap then overlap, interacting cash flo

Table 6: Benchmark configuration summary. N: number of projects; D_i: planned duration; s_i: planned start (episode period); M_i: non-advance milestones; ϕ_i: advance rate; ψ_i^{trig}: progress level at which the advance is disbursed (0.00 means at project start).

TC-1LMAIF-R — Delayed start and progress-triggered advance. One project, planned start $s_i = 2$, planned duration $D_i = 3$ (four active periods: $\tau_i \in \{0, 1, 2, 3\}$), three non-advance milestones, advance rate $\phi_i = 0.20$ with trigger at $\psi_i^{\text{trig}} = 0.80$. The trigger means the advance of $\phi_i \times P_i = 24$ is disbursed only once cumulative progress $EV_{i,t} \geq 0.80$ is reached — not at project start. Advance recovery rate $\rho_i = 0.50$ is the highest in the benchmark suite.

With milestone gross payments of 18, 42, and 60 (weights 0.15, 0.35, 0.50), the recovery trace is: $j = 1$: $\min(0.50 \times 18, 24) = 9.0$ recovered, advance remaining = 15.0; $j = 2$: $\min(0.50 \times 42, 15.0) = 15.0$ recovered, advance fully cleared; $j = 3$: no further recovery. Retention deductions (0.9, 2.1, 3.0) accumulate across the three milestones and are released in full (6.0) at $j = 3$. Net payments after all deductions: 8.1, 24.9, and 63.0. Efficiency sequence: [0.91, 1.17, 0.83, 0.91, 1.17, 0.83]. This configuration validates: (i) delayed project start and its interaction with the episode clock; (ii) progress-triggered advance disbursement; (iii) the liquidity trough created by high recovery deductions after the advance fires; and (iv) the deduction ordering invariant that advance recovery precedes retention on every gross payment.

TC-10MAIF-R — Diminishing efficiency and termination risk. One project, planned start $s_i = 0$, planned duration $D_i = 3$, three milestones, advance rate $\phi_i = 0.20$ disbursed at project start, advance recovery rate $\rho_i = 0.20$. The efficiency sequence is monotonically decreasing: [0.95, 0.91, 0.55, 0.40, 0.30, 0.20]. Early periods are productive; late periods are near-inoperable. The advance of 24 is disbursed at $t = 0$.

Milestone payment trace (gross payments 18, 42, 60): $j = 1$: advance recovered = $\min(0.20 \times 18, 24) = 3.6$, advance remaining = 20.4, retention = 0.9, net = 13.5; $j = 2$: recovered = $\min(0.20 \times 42, 20.4) = 8.4$, advance remaining = 12.0, retention = 2.1, net = 31.5; $j = 3$: recovered = 12.0 (advance fully cleared), retention = 3.0, released = 6.0, net = 51.0. This configuration stress-tests the termination mechanic: the sharp efficiency drop beyond $\tau_i = 1$ makes it progressively more expensive to keep $\delta_{i,t} \leq \delta_i^{\max} = 0.15$, and any policy that does not front-load work risks activating the cure-period counter in later periods.

TC-3XMAIF-R — Multi-project with gap and overlap. Three projects with heterogeneous scale ($\text{BAC}_i = [100, 200, 300]$, prices [120, 240, 360]) and staggered starts ($s_i = [0, 6, 7]$). Each project has planned duration $D_i = 3$, three milestones, and advance rate $\phi_i = 0.20$ disbursed at project start. Advances at episode periods 0, 6, and 7 are 24, 48, and 72 respectively, creating three distinct outflow spikes the policy must navigate against a shared budget.

The episode has two structurally distinct phases. *Gap phase* ($t = 4, 5$): Project 0 completes at $t = 3$; Projects 1 and 2 have not yet started. No allocations are required and the budget accumulates. *Overlap phase* ($t = 7, 8, 9$): Projects 1 and 2 are simultaneously active, creating competing allocation demands against the shared reserve B_t. The concurrent advance outflows at $t = 6$ (-48) and $t = 7$ (-72) produce the largest single-period deficit in the benchmark suite.

Per-project efficiency sequences: Project 0: [0.91, 1.17, 0.83, 0.91, 1.17, 0.83]; Project 1: [1.09, 0.84, 1.18, 1.09, 0.84, 1.18]; Project 2: [0.97, 1.13, 0.86, 0.97, 1.13, 0.86]. None is monotonically degrading; the challenge here is purely coordination and cash-flow timing, not individual project difficulty.

5.3 Solvers

Four methods are evaluated, defined formally in Section 3.5.5.

Reach-Plan (π^{rp}) allocates $x_{i,t}^{\text{rp}}$ to bring every active project exactly to its S-curve planned progress each period, regardless of budget scarcity or milestone proximity.

Catchup-T (π^{cu}) allocates the minimum spend $x_{i,t}^{\text{cu}}$ required to keep the progress delay within $\delta_i^{\max}$ this period, preserving budget for later periods and other projects.

Catchup-Next-T ($\pi^{\text{cu}+1}$) pre-emptively allocates $x_{i,t}^{\text{cu}+1}$ so that the projected delay next period also remains within $\delta_i^{\max}$, at the cost of higher early expenditure.

Full-information optimum (π^*) is computed manually with complete prior knowledge of all efficiency realisations ($\eta_{i,t}$), via the interactive `play.py` interface with the efficiency sequences fixed in advance. It serves as the upper-bound reference Z^* from Proposition 1. Implementing a tractable algorithmic solver (LP or DP) is left as future work; the manual computation suffices to establish the performance ceiling across the four configurations studied here.

All four methods are evaluated under seed = 0 with the fixed custom-ordinal efficiency sequences specified above, ensuring exact reproducibility across runs and solvers.

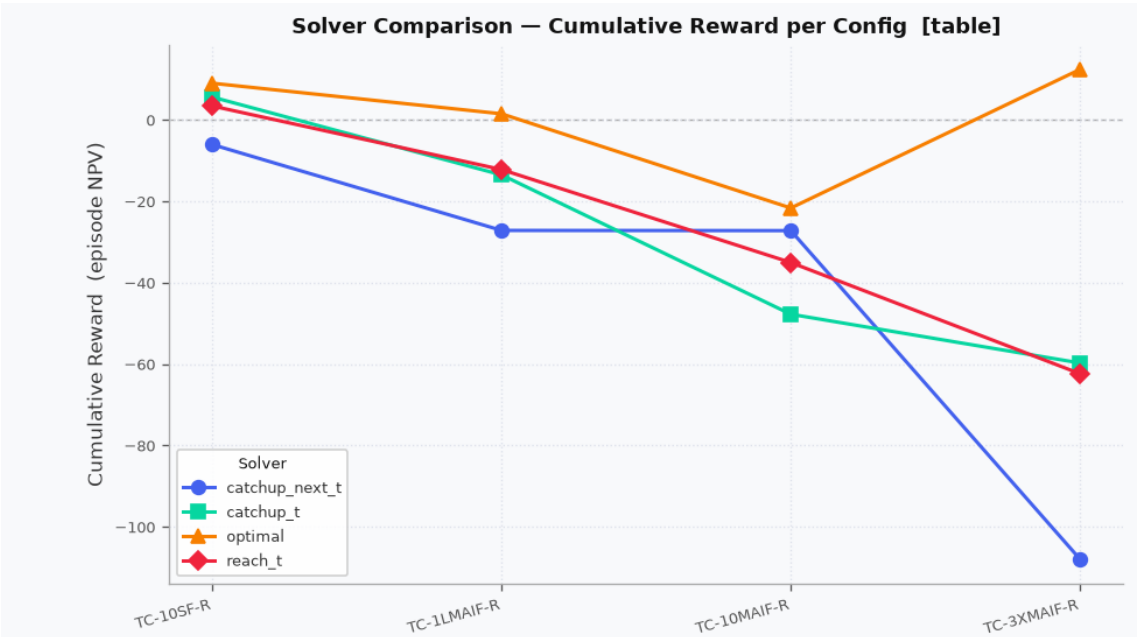
5.4 Results

Figure 7 reports cumulative episode NPV $G = \sum_{t=0}^{T} \gamma^t \Delta B_t$ for each method across the four configurations, ordered by increasing structural complexity. Table 7 presents the numerical values and the best-heuristic gap $Z^* - \max_\pi G(\pi)$.

Config	$G(\pi^*)$	$G(\pi^{\text{rp}})$	$G(\pi^{\text{cu}})$	$G(\pi^{\text{cu}+1})$	**Best-heuristic gap**
TC-10SF-R	9.00	5.60	3.40	-6.07	3.40
TC-1LMAIF-R	1.49	-13.21	-13.21	-27.17	14.70
TC-10MAIF-R	-21.66	-35.06	-47.77	-27.22	5.56
TC-3XMAIF-R	12.40	-59.75	-62.38	-107.96	72.15

Table 7: Episode NPV G for all methods across four configurations. Best-heuristic gap $= G(\pi^*) - \max\{G(\pi^{\text{rp}}), G(\pi^{\text{cu}}), G(\pi^{\text{cu}+1})\}$.

TC-10SF-R. On the isolated single-period configuration all three heuristics produce positive NPV and the gap is modest. Reach-Plan achieves 5.60 against the optimum 9.00 (gap 3.40); Catchup-T follows at 3.40. Catchup-Next-T finishes at -6.07: with only one active period and one milestone, its one-period lookahead over-spends relative to what



	10SF-R	1LMAIF-R	10MAIF-R	3XMAIF-R
catchup_next_t	-6.07	-27.17	-27.22	-107.96
catchup_t	5.60	-13.53	-47.77	-59.75
optimal	9.00	1.49	-21.66	12.40
reach_t	3.40	-12.21	-35.06	-62.38

Figure 7: Cumulative episode NPV G for three deterministic greedy heuristics and the full-information optimum across four benchmark configurations ordered by increasing portfolio complexity. The widening gap confirms that rule-based allocation deteriorates systematically as structural complexity grows.

the milestone requires, incurring interest costs without a compensating return. The result confirms that the environment correctly prices conservative over-allocation as a net loss.

TC-1LMAIF-R. The progress-triggered advance and high recovery rate ($\rho_i = 0.50$) create a liquidity trough immediately after the advance fires. All three heuristics produce negative NPV while the optimum remains marginally positive at 1.49; the best-heuristic gap is 14.70. The full-information optimum exploits the known efficiency sequence to time milestone certifications so that the advance trigger fires at the most advantageous moment and the large recovery deductions at $j = 1$ (-9.0) and $j = 2$ (-15.0) are offset by subsequent cash inflows. The greedy policies, allocating from current-period signals only, cannot anticipate the trigger timing and over-allocate during the low-return period before the advance fires. Both π^{rp} and π^{cu} achieve identical NPV (-13.21), indicating that the binding constraint is not the delay cap but payment timing: both rules satisfy the tolerance in the same periods, but neither correctly sequences allocation to capture the advance inflow early.

TC-10MAIF-R. The monotonically diminishing efficiency sequence makes the allocation timing problem acute. Even the full-information optimum achieves only -21.66: the late-stage efficiency collapse ($\eta_{i,t} = 0.30, 0.20$) makes full project recovery uneconomical. Catchup-Next-T (-27.22) outperforms both Reach-Plan (-35.06) and Catchup-T (-47.77): its risk-averse lookahead front-loads work into the high-efficiency early periods, reducing expenditure in the near-inoperable late periods where each unit of allocation yields little progress. Catchup-T performs worst because its minimum-spend rule continues allocating into low-efficiency periods to maintain the tolerance floor, accumulating cost without meaningful progress. The best-heuristic gap of 5.56 is the smallest in the suite, because the challenge here is project-level efficiency timing rather than multi-project coordination.

TC-3XMAIF-R. The multi-project configuration produces the starkest result. The full-information optimum achieves $+12.40$: with known efficiency sequences, the oracle times advance disbursements, milestone certifications, and the gap-phase budget accumulation to convert the complex portfolio into a profitable episode. Every greedy heuristic produces a loss — Reach-Plan at -59.75, Catchup-T at -62.38, Catchup-Next-T at -107.96 — and the best-heuristic gap reaches 72.15, a $21\times$ increase over TC-10SF-R. The sign reversal (profitable optimum, loss-making heuristics) is the central observation. The concurrent advance outflows at $t = 6$ (-48) and $t = 7$ (-72) create a large shared deficit that each single-project greedy rule handles independently, without accounting for how its own allocation depletes the reserve available to the concurrent project. Catchup-Next-T collapses to -107.96 because its aggressive pre-emptive spend across all three simultaneously-active projects exhausts the budget during the overlap phase, leaving insufficient reserve to certify the high-value late milestones of Projects 1 and 2.

5.5 Discussion

Three findings emerge from the benchmark study.

The gap is monotonically increasing with structural complexity. The best-heuristic gap grows from 3.40 (TC-10SF-R) to 14.70 (TC-1LMAIF-R) to 72.15 (TC-3XMAIF-R), a $21\times$ increase from simplest to most complex. TC-10MAIF-R (5.56) is

the only departure from strict monotonicity: its challenge is project-level efficiency timing, not multi-project coordination, so the gap is smaller than TC-1LMAIF-R's despite higher individual-project difficulty. The gap in TC-3XMAIF-R is not marginal degradation; it reflects a structural failure of greedy allocation under shared-budget coordination requirements that no single-project rule can resolve.

No single heuristic dominates. Each heuristic is best on at least one configuration: Reach-Plan on TC-10SF-R, Catchup-Next-T on TC-10MAIF-R, Catchup-T on TC-3XMAIF-R. This confirms that the optimal policy is adaptive: the correct allocation depends jointly on the efficiency trajectory, the milestone payment structure, the advance trigger timing, and the phasing of concurrent projects — a combination no fixed rule can capture. The failure mode of each heuristic is traceable to a specific structural feature: Reach-Plan over-spends in aggregate; Catchup-T under-anticipates the lookahead; Catchup-Next-T over-commits in the multi-project overlap phase.

A well-defined improvement target exists for learned agents. The full-information optimum $G(\pi^*)$ provides a concrete upper bound (Proposition 1) for each configuration. On TC-3XMAIF-R the swing between the best achievable (+12.40) and the best greedy (−59.75) is 72.15 NPV units — the difference between a profitable and a deeply loss-making portfolio outcome. The environment mechanics are thereby empirically validated: each configuration produces the expected pattern of cash-flow events (advance disbursements, milestone certifications, recovery deductions, retention release, interest accrual) correctly sequenced and financially consistent, and the full-information optimum exploits structural features that greedy rules are architecturally unable to access. This gap is the value of intelligent sequential allocation and constitutes the performance target for future RL agents trained on the environment of Section 3.

6 Conclusion

7 Conclusion

7.1 Summary of Contributions

This paper addresses a structural gap in the project portfolio management literature: the absence of a formal, computationally tractable framework for operational budget allocation across a committed portfolio of FIDIC-governed projects under cash-flow uncertainty.

Three contributions are delivered.

C1 Gymnasium-compatible MDP environment. `PortfolioBudgetingEnv` is a fully parameterisable, open-source Gymnasium environment encoding the complete FIDIC cash-flow cycle: advance payment and progressive recovery, progress-weighted milestone payments with configurable weights and timestep eligibility constraints, retention withholding and release, treasury-draw interest on project cash deficits, and a two-condition termination gate with a project-specific cure-period counter. The sole stochastic primitive is the budget-to-progress efficiency $\eta_{i,t}$, drawn each period from a configurable distribution. All parameters — schedule, contract, and portfolio — are specified through a configuration dictionary whose entries are probability distributions, enabling domain randomisation and calibration to any project-based sector without code modification. Gymnasium API compliance is verified across 51 checks in 10 test sections.

A SQLite audit ledger records every allocation, cash-flow event, breach flag, and EVM index at each step, making policy behaviour fully interpretable.

C2 Formal MDP specification. The portfolio budget allocation problem is formalised as a finite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma, T)$ with state dimension $9N + 1$, a $7N + 2$ dimensional observation vector derived from actionable signals rather than raw EVM indices, a three-step action projection guaranteeing feasibility for any policy output, and a reward function equal to the discounted net portfolio cash flow. All structural incentives — completion, breach avoidance, milestone timing, and treasury-draw deterrence — emerge from the financial mechanics without auxiliary penalty terms. The Markov property holds exactly: the full portfolio state S_t is a sufficient statistic for the observable history.

C3 Benchmark evaluation and quantified optimality gap. Three deterministic greedy heuristics (Reach-Plan, Catchup-T, Catchup-Next-T) are evaluated against a manually computed full-information optimum across four benchmark configurations of increasing structural complexity, with fixed efficiency sequences ensuring exact reproducibility. The best-heuristic optimality gap grows from 3.40 NPV units on the isolated single-project atom to 72.15 units on the three-project coordinated portfolio — a 21× increase. On the most complex configuration, every greedy heuristic produces a loss while the full-information optimum remains profitable, establishing both the validity of the environment mechanics and a concrete performance target for future learned agents.

7.2 Implications

For sequential decision-making research. The environment provides a problem instance that is simultaneously realistic (FIDIC mechanics calibrated to empirical construction practice), formally specified (closed-form MDP with exact Markov property), and computationally tractable (polynomial step cost, episode length $T \sim$ 20–30 periods for representative portfolios). The intractability of exact stochastic optimisation — K^{NT} scenario-tree leaves, $K^{72} \approx 10^{50}$ for a moderate three-project instance — makes this a well-motivated benchmark for RL and approximate dynamic programming methods. The full-information optimum provides an exact upper bound (Proposition 1) against which any policy can be evaluated without requiring a solver.

For project management practice. The experiment results show that no fixed rule performs consistently well: the optimal policy is adaptive to the efficiency trajectory, the milestone payment structure, the advance trigger timing, and the phasing of concurrent projects simultaneously. For a portfolio manager, this implies that rule-of-thumb allocation — even well-calibrated rules such as Catchup-T, which tracks the contractual tolerance floor — leaves substantial value on the table as portfolio complexity grows. The audit ledger provides a deployment-ready interpretability layer: every allocation, payment, and breach event is recorded per period, making the environment suitable as a decision-support tool even before a learned policy is trained.

For domain randomisation and generalisation. Each parameter in the configuration dictionary can be specified as a probability distribution, and N (portfolio size) can itself be a configurable random variable. This design supports episode-level domain randomisation (Tobin et al., 2017) natively, without any modification to the environment code. A policy trained across a distribution of configurations is exposed to diverse advance structures, efficiency profiles, milestone weights, and portfolio sizes

within each training run, a prerequisite for generalisation across the range of contract types and market conditions encountered in practice.

7.3 Limitations

The present formulation makes three deliberate simplifications, each of which is a natural boundary for future extension.

Independent projects. Projects interact only through the shared budget B_t. Shared workforce or equipment pools, technology spillovers, and client-side interdependencies are not modelled. Extending to resource-coupled projects would require augmenting the state space with resource inventories and adding coupling constraints to the action projection.

Milestone payments only. The payment structure follows FIDIC milestone-certification mechanics. Time-based, cost-plus, and hybrid contract forms are not currently supported, though the configuration dictionary architecture accommodates them through additional milestone profile parameters.

Single stochastic primitive. The budget-to-progress efficiency $\eta_{i,t}$ absorbs all uncertainty. Scope changes, external price shocks, and client payment delays are exogenous. This restriction cleanly identifies the allocation problem structure; a multi-source uncertainty extension would require additional state components to maintain the Markov property.

7.4 Future Work

The environment and benchmark established here define a clear research agenda.

RL agent training. The immediate next step is to train a policy — PPO (Schulman et al., 2017) is a natural starting point given its stability in continuous action spaces — on a domain-randomised distribution of configurations and evaluate it against the benchmark configurations of Section 5. The full-information gap of Table 7 provides the evaluation target; a trained policy that closes a substantial fraction of the TC-3XMAIF-R gap (72.15 units) would constitute a meaningful improvement over all greedy baselines.

LLM-augmented policies. The observation vector o_t is structured, low-dimensional ($7N + 2$ components), and semantically interpretable — each component has a direct financial or contractual meaning. This makes it a natural input for large language model-based reasoning agents integrated via LangChain or similar frameworks, with the benchmark heuristics serving as reward signals for in-context action scoring. The primary challenge for such an approach is generalisation across configurations: a policy that overfits to the four benchmark efficiency sequences does not transfer to novel portfolios. Generalisation configurations designed to inject controlled uncertainty — varied S-curve shapes, heterogeneous milestone weights, mixed advance structures — are the appropriate next step for evaluating policy robustness.

Formal algorithmic upper bound. The manually computed Z^* is exact for the four fixed-efficiency configurations but does not extend to stochastic settings. A Linear Programming relaxation of the deterministic oracle problem (19), or a Dynamic Programming solution for small instances, would provide a tighter and algorithmically verifiable bound against which trained policies can be compared.

Real-world deployment. Integration with Oracle Primavera P6 or equivalent portfolio management information systems would enable deployment as a decision-support tool, with live EVM exports providing the state vector and the environment serving as a scenario simulator for prospective allocation decisions. The SQLite audit ledger is

structurally compatible with standard EVM reporting schemas and requires no additional instrumentation.

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A Appendix A : PRISMA 2020 Flow Diagram

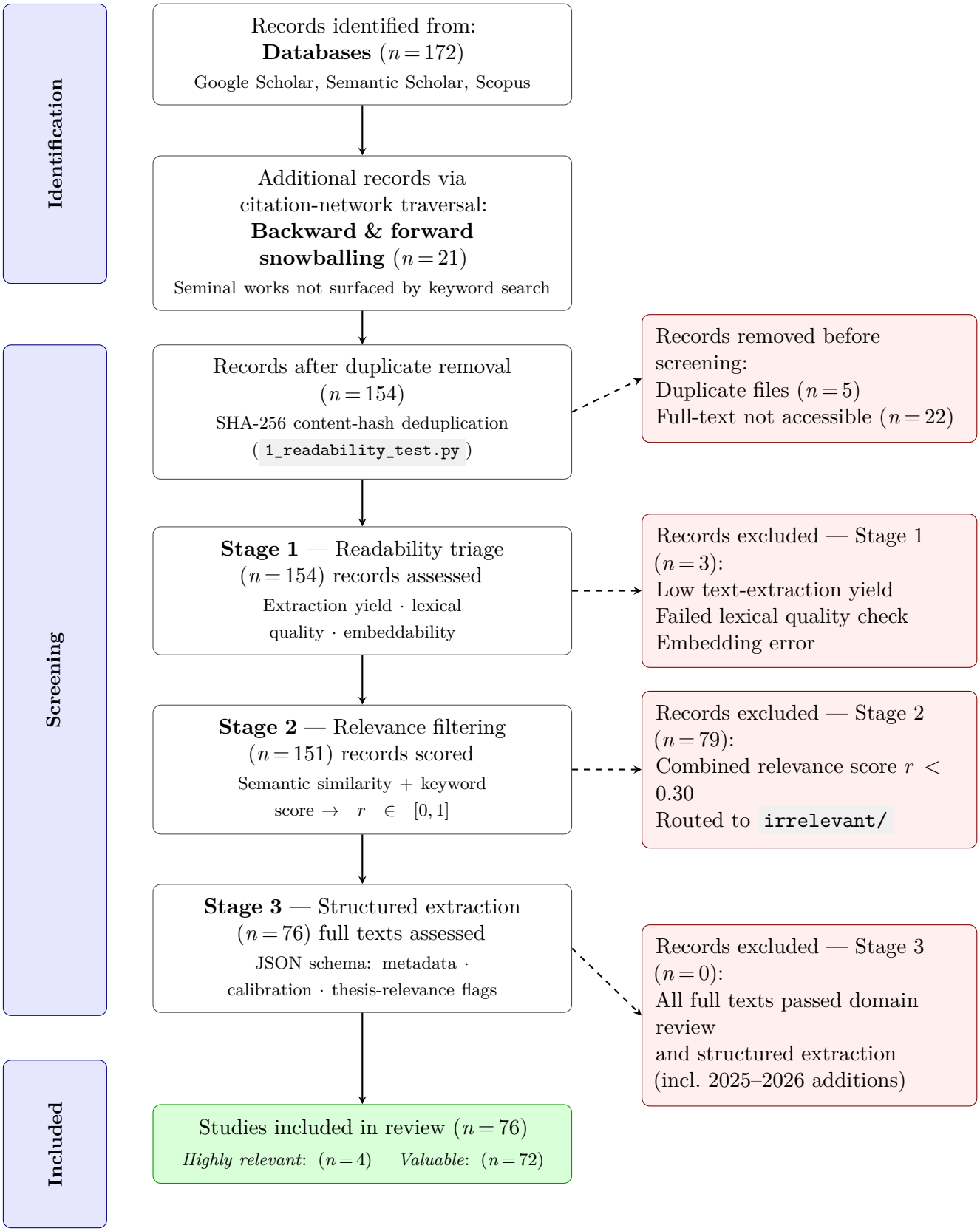


Figure 8: PRISMA 2020 flow diagram of the automated literature search and screening process. Stage labels correspond to scripts in `docs/lit_archive/`.

B Appendix B : Gymnasium Compatibility Certificate

`tests/gym_test.py` exercises `PortfolioBudgetingEnv` across ten test sections and 51 individual checks, covering import and instantiation, space declarations, the `reset()` and `step()` API contracts, determinism under fixed and varying seeds, random-policy and degenerate (zero-action, greedy) rollouts, multi-episode stability, and the official `gymnasium.utils.check_env` checker. The suite is run against benchmark configuration `TC-3XMAIF-R` (3 projects, mixed start times, fixed efficiency sequences). Table 8 summarises the ten sections; all 51 checks pass.

Table 8: Gymnasium compatibility test summary — `PortfolioBudgetingEnv`, config `TC-3XMAIF-R`, generated 2026-10-20.

§	Section	Passed
1	Import & Instantiation	8 / 8
2	Space Declarations	7 / 7
3	`reset()` Contract	13 / 13
4	`step()` Contract	10 / 10
5	Determinism	2 / 2
6	Random Policy Rollout	6 / 6
7	Zero-Action Rollout	1 / 1
8	Greedy Rollout	1 / 1
9	Multi-Episode Stability	2 / 2
10	`gymnasium.utils.env_checker`	1 / 1
Total		**51 / 51**

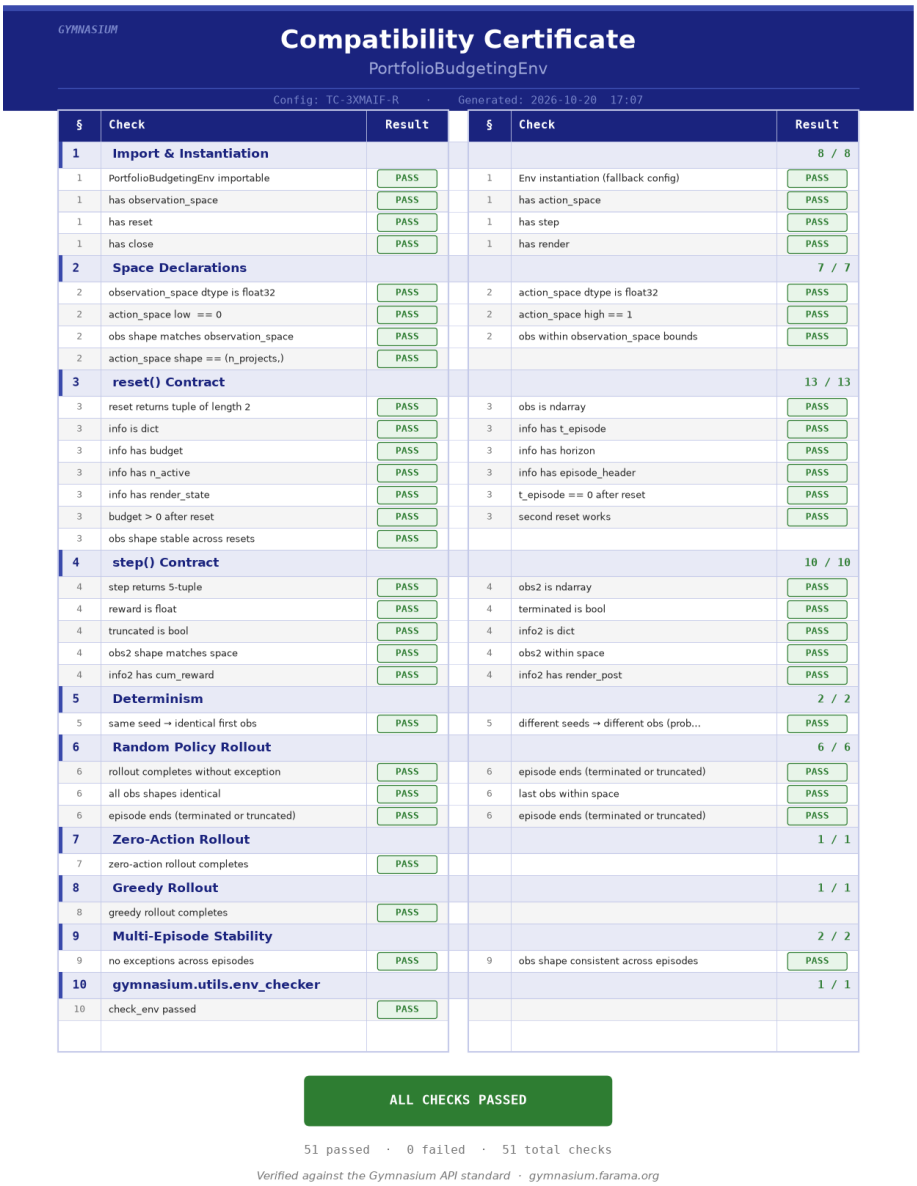


Figure 9: Compatibility certificate generated by `tests/gym_test.py`. All 51 checks pass; 0 failures.

C Reopsitory Architecture and Concept Map

C.1 Repository Architecture Overview

The complete implementation accompanying this paper — the environment, the baseline solvers, the verification suite, and the literature-review pipeline described in Section 2 — is organised as a single version-controlled repository. The repository separates *documentation* (this manuscript, the thesis, and the literature corpus) from *implementation* (the environment, ledger, baselines, and tests), as summarised below.

```
ppm_env/
|-- notation.yaml        parameter dictionary (academic <-> code)
|-- docs/
|   |-- article/          LaTeX source of this manuscript
|   |-- lit_archive/      literature screening pipeline and corpus
|   |-- mscn/             supplementary thesis documentation
|   |-- pp_presentation/  thesis defense slides (university deliverable)
|   `-- thesis_farsi/     Persian thesis manuscript (university deliverable)
`-- src/
    |-- db/               ledger schema, configs, run records
    |-- env/              core CMDP environment implementation
    `-- tests/            verification suite, baselines, gap analysis
```

The `mscn/`, `pp_presentation/`, and `thesis_farsi/` directories satisfy institutional degree-completion requirements and fall outside the reproducibility scope of this paper; they are listed here only for completeness. The `notation.yaml` file at the repository root provides an explicit, machine-readable correspondence between every academic symbol introduced in Section 3 and its counterpart variable name in the source code, and is intended as the primary cross-reference for readers wishing to connect the formal model to the implementation.

The literature-review pipeline described in Section 2.1.1 resides in `docs/lit_archive/` and retains, alongside the processing scripts, the full PDF archive (`lit_archive_passed/`, `lit_archive_failed/`) and the structured JSON extraction records that drove the synthesis table and gap-analysis figures reported in Section 2.

C.2 Core Contribution Modules

The modules implementing the formal model of Section 3 and the evaluation of Section 5 are contained entirely within `src/`. The environment dynamics — state construction, EVM/S-curve computation, the FIDIC cash-flow cycle, and the termination gate — are implemented in `src/env/`, with the environment's public Gymnasium-compatible interface and its internal support routines maintained as separate modules within that directory. During training or evaluation, the environment class is instantiated per episode through the standard Gymnasium `reset()` / `step()` interface; no additional driver component is required.

The SQLite audit ledger introduced in Section 3 is schema-defined and persisted under `src/db/`, which also holds the benchmark portfolio configurations used in Section 5 and the run-record databases generated by each executed episode.

The verification and evaluation layer resides in `src/tests/`, which bundles four functionally distinct components: the Gymnasium compliance suite, the baseline and optimal solvers, the benchmark configuration generator together with its archived reference episodes, and the post-hoc analysis scripts used to produce the figures and tables of Section 5.

```
src/tests/
|-- gym_test.py                 Gymnasium API compliance suite
|-- gym_certificate.png         compliance certificate (Appendix B)
|-- baseline_solvers.py         greedy heuristics and optimum solver
|-- reference_episodes_gen/     benchmark configuration generator
|-- reference_episodes_record/  archived configs and solutions
`-- analysis/                   gap-analysis and figure scripts
```

The `reference_episodes_record/` directory retains the configuration set and recorded solutions used to validate the environment and to produce the four benchmark configurations evaluated in Section 5; it is intended as a working template for practitioners who wish to instantiate the environment against their own company data.

C.3 Concept-to-Implementation Traceability Matrix

Table 9 maps every concept and promised contribution named in the main text to its corresponding location in the repository, and is provided to support independent verification of the claims made throughout this paper.

Table 9: Traceability matrix mapping concepts and promised contributions to their implementation in the repository.

Concept / Contribution	Ref.	Repository Location	Role
Academic–code notation dictionary	§3	`notation.yaml`	Maps formal symbols to code identifiers
EVM/S-curve state representation	§3	`src/env/`	State vector construction
FIDIC cash-flow transition dynamics	§3	`src/env/`	Advance, retention, interest, and milestone logic
Two-condition termination gate	§3	`src/env/`	Episode termination evaluation
SQLite audit ledger	§3	`src/db/`	Schema definition and per-run event logging
Benchmark portfolio configurations	§5	`src/db/`	Stored configuration instances for evaluation runs
Gymnasium compliance verification	App. B	`src/tests/`	API conformance tests and certificate generation
Greedy heuristic baselines (Reach-Plan, Catchup-T, Catchup-Next-T)	§5	`src/tests/`	Deterministic baseline policies

continued on next page

Table 9 – continued

Concept / Contribution	Ref.	Repository Location	Role
Manually computed full-information optimum	§5	`src/tests/`	Upper-bound benchmark solver
Benchmark configuration generation and reference episodes	§5	`src/tests/`	Configuration matrix, episode archive, and company-data templates
Gap-analysis figures and dashboards	§5	`src/tests/`	Post-hoc analysis and figure generation
Literature screening, Stages 1–3	§2.1.1	`docs/lit_archive/`	Readability triage, relevance filtering, structured extraction
Synthesis dashboard and literature map	§2.1.1	`docs/lit_archive/`	Gap matrix, domain charts, synthesis table generation
Manuscript source	—	`docs/article/`	LaTeX source, figures, and tables for this paper

C.4 Reproducibility Notes

The mapping in Table 9 reflects a specific snapshot of the repository at the time of writing; subsequent development may reorganise individual modules without affecting the directory-level structure described in this appendix. Access to the repository, including the full commit history, the benchmark configurations, and the Gymnasium compliance certificate reproduced in Appendix B, is available from the corresponding author upon request.